CHAPTER

6

Group Reporting V

Equity Accounting under IAS 28 Joint Arrangements under IFRS 11



LEARNING OBJECTIVES

After reading this chapter, you should be able to:

LO1 Understand the concept of “significant influence” and “joint control”;

LO2 Appreciate the different accounting policies for investment in an associate and joint venture as reflected in an investor’s separate financial statements and the consolidated financial statements;

LO3 Understand the differences between the cost and equity method and consolidation;

LO4 Know how to apply the equity method in accounting for investment in an associate and joint venture in the consolidated financial statements;

LO5 Perform an analytical check on the balance of the investment in an associate and joint venture; and

LO6 Appreciate the nature of joint arrangements and know how to apply the appropriate accounting treatment.

EQUITY METHOD VERSUS COST METHOD

Concept of "Significant Influence"

Chapter 2 explained the concepts of "significant influence" and "associate." Significant influence is the power to participate in the financial and operating policy decisions of the investee, but is not control or joint control of those policies. An investor of an associate is not a "parent," as defined by IFRS 10 *Consolidated Financial Statements*. In IFRS 10, a parent is an entity that "controls" another entity (that is, a subsidiary). We saw in Chapter 2 that an investor has control over an investee when that investor has an exposure or rights to variable returns of the investee and has the ability to use its power to affect those returns.[1] In other words, three important elements exist in a control relationship. A control relationship exists when the investor has power over the investee. However, the investor must also have the ability to use the power. For example, if an investor has power through voting rights, but the investee's business is expropriated by some higher authority, the investor is quite helpless and the power through voting rights is ineffective. Finally, the investor must be exposed or have rights to variable returns of the investee. We show the 3-element control relationship in Figure 6.1.

FIGURE 6.1 Control



Joint control, on the other hand, is the contractual sharing of control, which requires all parties in a joint arrangement to unanimously agree on decisions about relevant activities of the arrangement.[2] For joint control to exist, two elements must be present (see Figure 6.2).

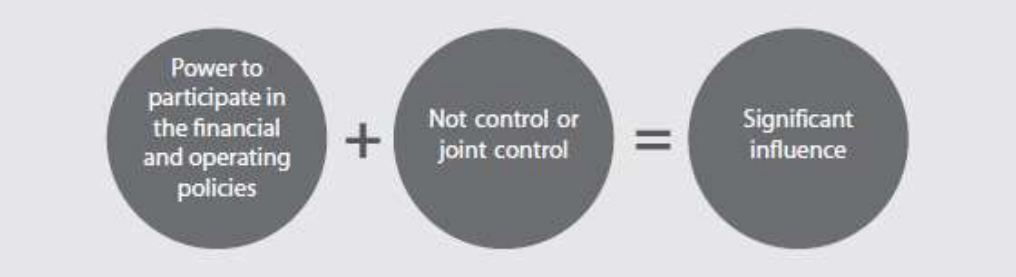
FIGURE 6.2 Joint control



By comparison, significant influence (see Figure 6.3) is not as powerful as control or joint control. It lacks the ability to determine the final decisions made on relevant activities of the investee. An investor's consent is also not required to arrive at these decisions when the investor has only significant influence. However, investors with

significant influence may participate in the process of decision-making. The existence of significant influence is evidenced in different ways. For example, representation on the board of directors is common evidence of significant influence. In some situations, an investor may have significant influence over an investee by virtue of a strategic relationship between the two parties. There may be material transactions between the investor and investee, interchange of managerial personnel, or provision of essential technical information.[3]

FIGURE 6.3 Significant influence



IAS 28 *Investments in Associates and Joint Ventures* sets a default presumption that the direct or indirect holding of 20% or more of the voting power of an investee leads to significant influence.[4] The upper limit is when control begins. As 20% is only a presumed minimum threshold for significant influence, an investor may depart from this threshold if the investor is able to demonstrate that the quantitative threshold is not indicative of significant influence. In the spirit of principles-based accounting standards of the International Accounting Standards Board (the Board), IAS 28 allows the investor to show that there is significant influence even when the direct or indirect holding falls below 20%. Conversely, the investor may also show that there is no significant influence when the direct or indirect holding is 20% or more. However, such departures from the default threshold must be explained and disclosed in the footnotes. The party that has significant influence over an "associate" is simply referred to as an investor in IAS 28. As with accounting for investments in subsidiaries, accounting for investments in associates differs with respect to the different levels of financial statements presented as shown in Table 6.1.

TABLE 6.1 Accounting policy for investments in associates

Levels of financial reporting	Accounting policy for accounting for investments in associates
1. Investor's separate financial statements	Cost or in accordance with IFRS 9 *Financial Instruments:* or the equity method
2. Consolidated financial statements: where an investor has investments in both subsidiaries and associates	Equity method
3. Investor's financial statements in place of consolidated financial statements: where an investor has investments in associates but not subsidiaries	Equity method

An investor of an associate must present the separate financial statements for itself as a legal entity, *and* either the consolidated financial statements or an investor's financial statements in place of consolidated financial statements. The last two financial statements relate to economic entity reporting. In some situations, the investor has only associates and will not present consolidated financial statements. However, the investor has to present two sets of

financial statements, one for the legal entity and the other for the economic entity, using the equity method. While the cost method is used often in reporting the investment in associates in the legal entity's financial statements, the equity method is used in reporting the investment in associates in the economic entity's financial statements. The investor is exempted from applying the equity method in the economic entity's financial statements in one of the following situations:

1. When the exemptions to consolidation in IFRS 10 applies. For example, the investor is a wholly-owned subsidiary of another investor and the investor's shares are not publicly traded. Chapter 3 explains the other exemptions.
2. When the investor is a venture capital organization, mutual fund of unit trust, or a similar organization, the investor may choose to measure the investment at fair value through profit or loss in accordance with IFRS 9 *Financial Instruments.*

Accounting for investments in associates in the investor's separate financial statements is the same as accounting for investments in subsidiaries in the investor's separate financial statements (IAS 27:10), that is, the investment may be measured either at cost or as a financial instrument in accordance with IFRS 9 *Financial Instruments*. For financial periods beginning on or after 1 January 2016, the equity method may be applied in accounting for investments in associates in the investor's separate financial statements (IAS 28:10). The same accounting method applies to all investments in each category.

What Is the Equity Method?

The **equity method** is "a method of accounting whereby the investment is initially recognized at cost and adjusted thereafter for the post-acquisition change in the investor's share of net assets of the investee. The profit or loss of the investor includes the investor's share of the profit or loss of the investee" (IAS 28:3).

In a nutshell, equity accounting essentially:

1. "Capitalizes" profits by adding them to the initial cost of the investment;
2. Does not eliminate the investment in associate account but presents the investment balance as the investor's share of the net assets of the associate including the unamortized balance of fair value adjustments, and the implicit goodwill held by the investor in an associate; and
3. Does not treat dividends as income but as "repayment" of the capitalized profits.

Contrast this with the *cost method* whereby the investment is recognized at initial cost, and the investor recognizes dividend income in profit or loss. The cost method is asymmetric; only impairment losses are recognized but no recognition is made for increases in the carrying amount of the investment.

Essentially, the equity method results in a carrying amount that is neither cost nor fair value but somewhere in between. The carrying amount of the investment under the equity method increases or decreases over time with the profits or losses made by an associate, and is reduced by the distribution of profits made by the associate. Dividends are deemed as a realization of profit rather than profit itself. The "capitalization" of profit increases the investment account and the retained earnings of the investor. The "capitalized" profits are an expectation of future dividend payments, and hence, increases the carrying amount of the asset. However, an investment carried under the equity method is not an approximation of fair value because equity accounting recognizes book profits and not the fair value changes of an investment.

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Although the equity method does not capture the fair value of an investment, it is a more timely method of recognizing profit from an associate than the cost method. Since dividends lag behind profit, the earlier recognition of the share of profit under the equity method provides more timely information about the performance of an associate company than the cost method. Earlier recognition of the share of profit of an associate also provides users of financial statements with a basis to predict future dividend cash flows from an associate.

Comparison between the Cost Method, Equity Method, and Fair Valuation

Although there appears to be significant differences between the cost, equity and fair value methods, the main difference is one of *timing*. Eventually, the total stream of income that flows from each method are the same. The cost method, although slower in recognizing profit, catches up with a large terminal profit on sale. Fair value through profit or loss (FVTPL), on the other hand, provides the most timely and dynamic form of income recognition. Table 6.2 shows a comparison of the three methods.

TABLE 6.2 Comparison between cost and equity methods and fair value through profit or loss (FVTPL)

Dimensions	Cost method	Equity method	FVTPL
Income recognition	Dividend income – approximates cash basis for income recognition, emphasizing reliability, and realized income	Share of profits approximates the accrual basis of income recognition, emphasizing the predictive value of information of unrealized gains	Dividend income and change in fair value providing timely information and emphasizing relevance over reliability
Asset measurement	Initial cost, less impairment losses	Initial cost and share of post-acquisition profit and other changes in equity	Fair value
Profit on sale	Large terminal profit on final realization	Smaller terminal profit as profits are recognized systematically over holding period, leaving a smaller terminal profit on sale	Zero terminal profit
Nature of the economic relationship between investor and investee	No economic relationship between investor and investee. Investor is deemed a passive holder of investments.	Economic interdependence between investor and associate – the economic entity comprises the investor and the associate through a relationship that is built upon significant influence	No economic relationship between investor and investee. The investment may be held for trading or qualifies for fair value measurement.

EQUITY METHOD VERSUS CONSOLIDATION

Rationale for Differences in Presentation

In the *consolidated financial statements* presented for the group as a single economic entity, the equity method must be applied to present the results and net assets of an associate. However, the investor need not apply the equity method when an investment is classified as held for sale in accordance with IFRS 5 *Non-current Assets Held for Sale and Discontinued Operations*.[5] Although the definition of a "group" excludes an associate (a group comprises a parent and its subsidiaries), a consolidated statement of financial position must measure the investment in an associate to capture the change in net assets of the associate. Similarly, the consolidated income statement must recognize the investor's share of the net profit of the associate.

Hence, the equity method measures the results of associates in a manner that effectively captures the substance of consolidation but not the form thereof. Unlike consolidation, the equity method does not combine the individual line items (for example, revenues and cost of sales) of the investor and the associate to arrive at consolidated totals for

each income statement or statement of financial position. Instead, the equity method recognizes the results of the associate through compressed single line items, "share of profit of associate" and "investment in associate."

Hence, the substance of consolidation is achieved through compressing the income statement items of an associate in these single measures. Instead of showing the profit item on a line-by-line basis and non-controlling interests as a deduction, the equity method shows a net figure that reflects only the investor's share in the profit of the associate. Similarly, the statement of financial position items are not consolidated, but the change in net assets of an associate is recognized in the investment in associate account. Effectively, consolidation and the equity method achieve the same group retained earnings and net assets.[6] However, the level of detail and information on revenues, expenses, assets, and liabilities are considerably lesser under the equity method because there is no aggregation of assets, liabilities, expenses, and revenues of the associate with the investor's.

Since the decision rights implicit in "significant influence" are not as strong as when "control" exists, the integration and interdependencies between investor and associate are likely to be less so than in parent-subsidiary relationships. Hence, it may be potentially misleading to show the revenues, expenses, and assets and liabilities of the investor and the associate as a single economic entity when the relationship is more distant. The equity method provides an accounting basis whereby changes in net assets of the associate are recognized in a timely manner but without the line-by-line aggregation of individual assets and liabilities of consolidation.

Financial Statement Differences between Consolidation and the Equity Method

Table 6.3 summarizes the differences between consolidation and the equity method. Essentially, the primary difference lies in the presentation and aggregation of information. The equity method "reveals" less in that an investor's share of the associate's profit and net assets is compressed into single line items.

TABLE 6.3 Comparison between consolidation and the equity method

Dimensions	Consolidation	Equity method
Income recognition	Each line item in the subsidiary's income statement is added in full to the parent's income statement. Hence, the consolidated income statement shows combined revenues and expenses of the parent and subsidiaries.	Share of profit is reported as a single line item in the consolidated income statement. Disaggregated information relating to revenues and expenses of associates are not included in the consolidated income statement.
Non-controlling interests	Since the revenues and expenses of subsidiaries are added in full to the parent's, non-controlling interests' share of profit after tax for the year is shown as an allocation of consolidated profit after tax	Equity accounting takes into account only the investor's share of profit. Hence, a separate line item for non-controlling interests is not necessary.
Asset measurement	Investments in subsidiaries are eliminated on consolidation. The investment account is substituted with line-by-line addition of the net assets of the subsidiaries, goodwill and fair value adjustments	Investments in associates are carried at initial cost plus share of post-acquisition profits and other changes in equity. The investor's share of the book value of the net assets of the associates is included in the investment account, together with share of fair value adjustments and the implicit goodwill. Through a single line item, the investment account shows the investor's

		interest in the associates' net assets at year-end.
Goodwill on consolidation	Unimpaired goodwill is shown separately as an asset on the consolidated statement of financial position	Unimpaired goodwill is implicit in the investment in associate account
Unamortized balance of fair value adjustments of identifiable net assets	Adjustments are made to the specific assets or liabilities on the consolidated statement of financial position	Unamortized balance of fair value adjustments of identifiable net assets is implicit in the investment in associate account
Profit on sale of the investment	Profit on sale = Sales proceeds – Initial cost + Share of post-acquisition change in equity	Profit on sale = Sales proceeds – Initial cost + Share of post-acquisition change in equity
Economic relationship between investor and investee	Economic integration between investor and investee based on "control" by investor	Relatively lesser degree of economic integration between investor and investee based on "significant influence" by investor
Impact on financial ratios	Because of the line-by-line summation, certain reported items (for example, sales or total liabilities) are larger under consolidation. Ratios such as the net profit margin and the debt-equity ratio are likely to be different under consolidation vis-a-vis the equity method.	As the equity method does not entail the aggregation of line items of an associate, certain ratios (for example, debt-equity ratio) may appear more favorable under the equity method

AN OVERVIEW OF THE METHODOLOGY OF EQUITY ACCOUNTING

The equity method embodies the following accounting procedures. These are illustrated and elaborated in subsequent sections.

1. Investment is initially recorded at cost.
2. The initial investment at cost comprises three elements:
   (a) Investor's *share of the book value* or carrying amounts of the net assets of the associate as reported in its legal entity financial statements;
   (b) Investor's *share of the fair value adjustments* of identifiable net assets of the associate;
   (c) *Goodwill.*
3. Hence, goodwill that is implicit in the cost of investment is written off when impaired. How-ever, the test of impairment is performed for the investment as a whole and not separately for goodwill.
4. *Any fair value adjustment* included in the cost of the investment is amortized or expensed off, and the amortization charges are adjusted against the investor's share of profit/(loss) in the period when the amortization takes place.
5. The investor's share of *past and current amortization of any fair value adjustment* is adjusted against the investment in the associate and opening retained earnings and share of profit.
6. The initial investment is adjusted thereafter for the *post-acquisition change in the investor's share of net assets* of the investee.

7. Post-acquisition change in the investor's share of net assets includes recognizing the *share of the profit/(loss)* of the investee in each reporting period between initial recognition and disposal date.
8. *Share of current profit of the associate* will be adjusted for the unrealized effects of asset transfers between investor and associate. Essentially: page 335
   (a) Unrealized profit at the end of a reporting period will result in a downward adjustment to share of current profit; and
   (b) Realized profit in current period relating to a transfer from a previous period will result in an upward adjustment to share of current profit.
9. *Distributions* (for example, dividends) reduce the carrying value of investments and are deemed as a repayment of profits. Dividends are not a basis for recognizing income. Since the share of profit of the associate is recognized in profit or loss of the investor, dividends should not be recognized as profit as well. Instead, dividends are credited to the investment in associate account as a realization of the capitalized profit.
10. *Other changes in equity* (for example, increase in revaluation reserves) are also recognized proportionately in accordance with the investor's interest. Share of current revaluation reserves are recognized in the statement of profit or loss and other comprehensive income.

To help form a mental model of the equity method, the investment in associate account below shows the constituent elements:

Investment in Associates (Group SFP)

Initial cost*	
+ Share of post-acquisition profits to beginning of current period	– Past impairment of investment –/+ Past amortization of fair value adjustments – Share of unrealized profit included in opening retained earnings of either the investor or associate
+ Share of profit of current period +/– Share of adjustments of realized profit/(loss) from upstream and downstream sales in previous period +/– Share of other changes in equity	– Dividends – Current impairment of investment –/+ Share of current amortization of fair value adjustments –/+ Share of adjustments of unrealized profit/(loss) from upstream and downstream sales in current period

**Initial cost = Share of book value of net assets (or equity) of associate at initial recognition + Share of excess of fair value over book value of identifiable net assets at initial recognition + Goodwill*

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PERFORMING AN ANALYTICAL CHECK ON THE INVESTMENT IN ASSOCIATE BALANCE

As we have seen, investment in associate includes three components:

1. Share of book value of net assets;
2. Share of fair value adjustments; and
3. Goodwill.

You would recall that these three components also feature in our analysis of non-controlling interests (when the acquirer measures non-controlling interests at fair value). Interestingly, when non-controlling interests have significant influence in an investee, the investment balance on the non-controlling interests' statement of financial position would include these three components as well.

Subsequent to initial recognition, each item in the investment in associate account increases or decreases one of the three components. Table 6.4 illustrates this point by decomposing the investment in associate into the three components:

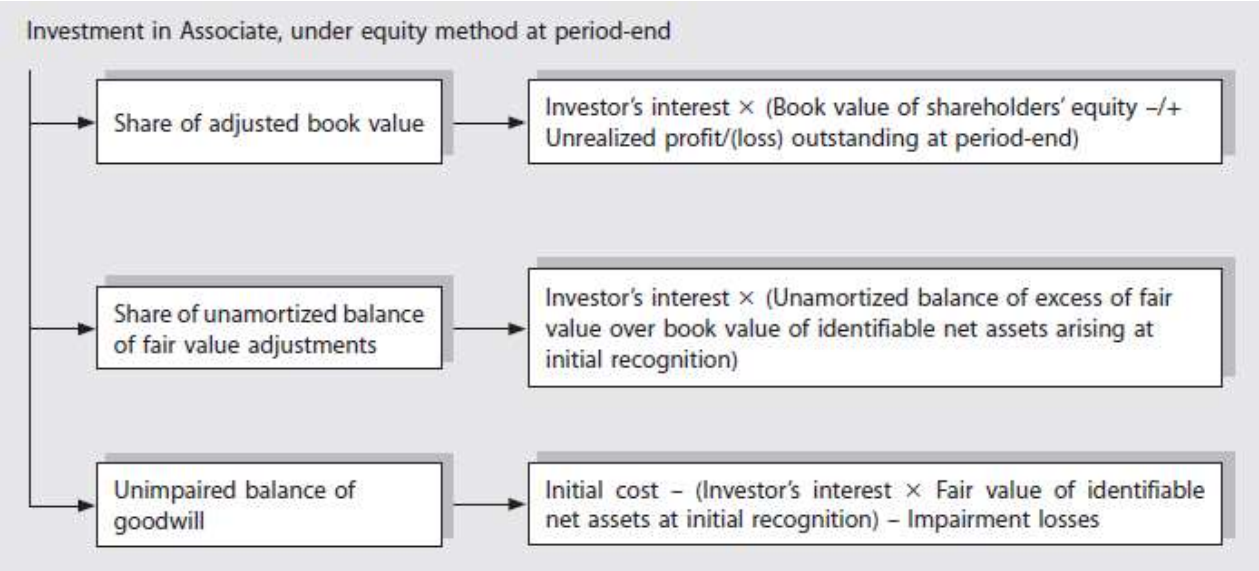
TABLE 6.4 Components of Investment in Associate

Period	Components of the Investment in Associate under the equity method		
At initial recognition	Share of book value of net assets or equity of associate at initial recognition	Share of fair value adjustments of identifiable net assets at initial recognition	Goodwill
From initial recognition to beginning of current period	+/− Share of post-acquisition profit/(loss) −/+ Share of adjustments of unrealized profit/(loss) in beginning retained earnings	−/+ Share of past amortization of fair value adjustments of identifiable net assets	− Past impairment of implicit goodwill
Current period	+/− Share of profit/(loss) after tax −/+ Share of adjustments of unrealized profit/(loss) from upstream and downstream sales in current period +/− Share of adjustments of realized profit/(loss) in current period in respect of previous period's upstream and downstream sales − Dividends +/− Share of other changes in equity	−/+ Share of current amortization of fair value adjustments of identifiable net assets	− Current impairment of goodwill[7]
Investment balance at end of current period	= Share of book value of net assets or equity of associate −/+ share of unrealized profit or loss at period-end = Share of *adjusted* book value of net assets or equity of associate at period-end	= Share of unamortized balance of fair value adjustments of identifiable net assets at period-end	= Unimpaired balance of goodwill at period-end

Hence, if we wish to check the period-end balance in the investment in associate, we can perform an analytical check by ascertaining the balance in these three components at the end of the year as shown in Figure 6.4.

Illustration 6.1 shows how the analytical check may be carried out.

FIGURE 6.4 Components of an analytical check on Investment in Associate

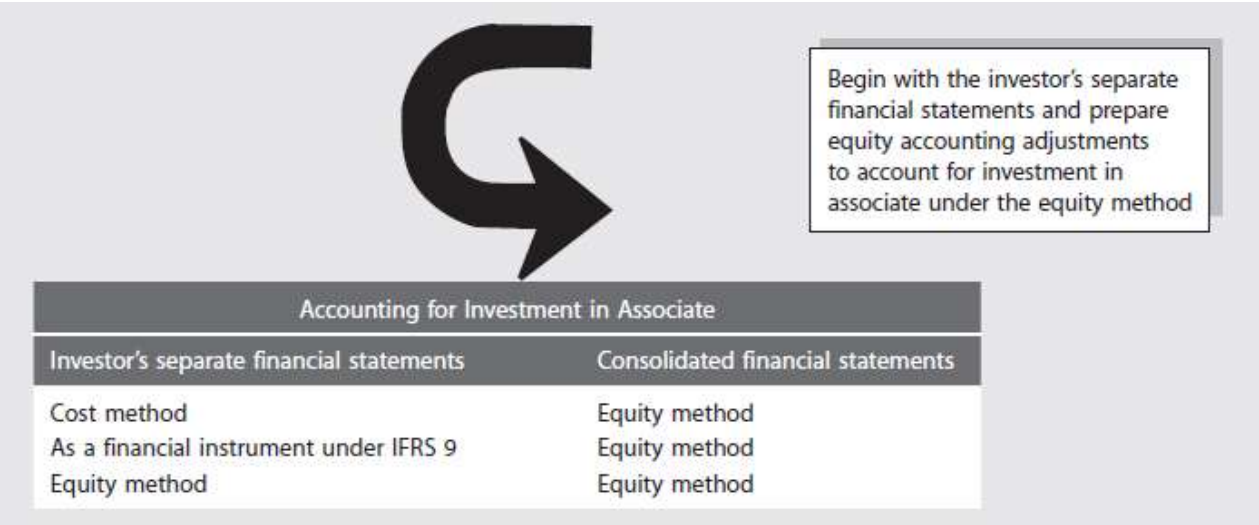


SPECIFIC PROCEDURES RELATING TO THE EQUITY METHOD

Conversion from the Cost Method to the Equity Method

Since we have to apply the equity method in the consolidated financial statements but the investment is normally carried at cost in the investor's separate financial statements, we have to make adjustments to the investment balance. The entries that we pass to effect the conversion to the equity method are known as "equity accounting entries." Adjustments are necessary to bring the basis of accounting from the cost method (or fair value) to the equity method (refer to Figure 6.5).

FIGURE 6.5 Conversion to the equity method



Accounting for Investment in Associate	
Investor's separate financial statements	Consolidated financial statements
Cost method	Equity method
As a financial instrument under IFRS 9	Equity method
Equity method	Equity method

In our illustrations, we assume that the investor carries the investment at cost. If the investment is carried at fair value under IFRS 9, additional equity accounting entries are required to strip off the fair value elements and bring the investment to its cost balance. If the investor uses the equity method to account for the investment in associate in the separate financial statements as permitted by IAS 27, no adjustments will be required to the consolidated financial statements.

IAS 28 paragraph 26 reinforces the underlying principle that the equity method results in the same substantive effects as consolidation:

> Many of the procedures that are appropriate for the application of the equity method are similar to the consolidation procedures described in IFRS 10. Furthermore, the concepts underlying the procedures used in accounting for the acquisition of a subsidiary are also adopted in accounting for the acquisition of an investment in an associate or joint venture.

Let us first consider some of the consolidation procedures that are not applicable to the equity method:

1. Because the equity method does not require line-by-line aggregation of financial statement items, the *elimination of balances* and *transactions between investor and associate are not required*.
2. *Investment in associate is not eliminated*. Instead, the investment captures multiple elements including implicit goodwill, changes in post-acquisition retained earnings of the associate, and the realization of earnings through dividends.

The consolidation procedures that apply to the equity method are discussed below.

Amortization of Fair Value Adjustments of Identifiable Net Assets

The investor must recognize *amortization of the undervaluation or overvaluation of identifiable assets and liabilities* that are included in the investment account. IAS 28 paragraph 32 requires appropriate adjustments to be made to the investor's share of the associate's profits or losses after acquisition to account for amortization of the associate's assets based on their fair values at initial recognition.

For example, I bought 20% of A's shares on 1 January 20x4 and had significant influence over A.

1. Initial investment in A was $6,000,000. I carries the investment at cost in its separate financial statements.
2. The excess of fair value over the carrying amount of a depreciable asset of A at 1 January 20x4 was $5,000,000. Depreciation was over a ten-year period.
3. Retained earnings of A on 1 January 20x4 was $15,000,000, while its retained earnings as at 1 January 20x5 was $20,000,000.
4. Net profit before tax of A for the current year ended 31 December 20x5 was $10,000,000 and its tax expense was $2,100,000 (there were some non-deductible items included in net profit).
5. Tax rate was 20%.

Dr Investment in associate	6,000,000	
Cr Cash		6,000,000

Initial investment in the associate

Explanatory note:

This entry is recorded in I's separate financial statements. The investment in associate is not eliminated and is included as an asset in the consolidated financial statements.

Dr Investment in associate	1,000,000	
Cr Opening retained earnings		1,000,000
Share of change in retained earnings from acquisition date to beginning of current period		

Retained earnings as at 1 January 20x5	$20,000,000
Retained earnings as at acquisition date	15,000,000
Change in retained earnings from acquisition date	$ 5,000,000
Share of A's post-acquisition retained earnings	$ 1,000,000

Explanatory note:

This entry re-enacts past equity accounting entries to arrive at the balance in the investment account under the equity method as at the end of the previous reporting year. By debiting the investment account and crediting retained earnings, this entry "capitalizes" the share of past profits in the investment account. The change in retained earnings captures the composite and cumulative effects of post-acquisition profits in past periods and the reducing effects of past post-acquisition dividend payments. This entry may be combined with the next entry on past depreciation of the undervalued fixed asset. Furthermore, if there are any unrealized profits in beginning retained earnings, an adjustment has to be made to the change in retained earnings.

Dr Opening retained earnings	100,000	
Cr Investment in associate		100,000
Share of past cumulative depreciation (20% × $5,000,000/10)		

Explanatory note:

Since equity accounting does not entail a line-by-line addition of assets and liabilities, any adjustment relating to an asset or liability of the associate is made against the investment in associate account. The investment account acts as a proxy for the "net assets" of the associate. In this example, the past cumulative depreciation relating to the undervalued fixed asset is debited to opening retained earnings to re-enact past entries relating to the depreciation. The credit entry is to the investment in associate to show that the net assets, which are represented by the investment in associate account, are reduced because of the cumulative depreciation effect. This entry may combine the tax effects of past amortization by applying the after-tax rate of 80% (refer to the composite entry below).

Dr Investment in associate	20,000	
Cr Opening retained earnings		20,000
Share of tax on past depreciation on undervalued fixed assets (20% × $100,000)		

Explanatory note:

Since past depreciation reduces past profits, tax expense in the past would also be reduced. The effect is to increase opening retained earnings and investment in associate. The reduction in tax expense enhances the investment in associate account. For simplicity, the tax adjustment entry can be combined with the depreciation entry to show the after-tax effect in the following composite entry.

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Dr Opening retained earnings	80,000

Cr Investment in associate		80,000

Share of the after-tax effects of past depreciation on undervalued fixed assets (20% × 80% × $5,000,000/10)

Dr Investment in associate	1,500,000	
Cr Share of profit of associate		1,500,000

Share of current profit after tax of associate (20% × $7,500,000)

The adjusted net profit after tax is as follows:

Net profit after tax	$7,900,000
Less current excess depreciation after tax	(400,000)
Adjusted net profit after tax	$7,500,000

Explanatory note:
Equity accounting recognizes the share of profit or loss for the period together with the share of tax of the associate. The split between profit and taxes is conceptually purer as tax expense is shown as a separate line item in the income statement. Investment income, such as dividend and interest, is also shown on a pre-tax basis. However, the *Guidance on Implementing Presentation of Financial Statements* presents the share of profit of associates after tax and non-controlling interests in the associates as a single line item. The *Guidance* is not a part of IAS 1 *Presentation of Financial Statements*, but in practice, most companies follow this presentation. In our illustrations and problems, we will present share of profit on an after-tax basis. Net profit after tax of the associate has to be adjusted for any amortization of fair value adjustments relating to the current period.

Goodwill Impairment

Since goodwill is not separately recognized in the equity method, it is included in the investment balance. Any impairment in goodwill will be credited to the investment in associate account. Impairment tests are explained in IAS 28 paragraph 42. Since goodwill included in the carrying amount of an investment in an associate is not recognized as a stand-alone asset but is implicit in the investment account, it is not tested for impairment on its own. Specifically under IAS 28, after application of the equity method where losses of the associate, if any are recognized, the entire carrying value of the investment for associate is tested for impairment in accordance with the requirements of IAS 36 *Impairment of Assets* when there is an indicator of impairment of the investment under IAS 28, that is, the guidance in IAS 28 is applied to determine if there is any objective evidence of impairment. If there is, the principles in IAS 36 will provide guidance on the measurement and recognition of any impairment losses. In particular, an impairment loss is recognized when the carrying value of the investment exceeds its recoverable amount. However, paragraph 42 of IAS 28 clarifies that such impairment loss is not allocated to any asset, including goodwill, that forms part of the carrying value of the investment in associate. If subsequent analysis reveals that the recoverable amount has increased, a reversal of the impairment loss is recognized in accordance with IAS 36 *Impairment of Assets*. Similarly, the accounting for the reversal of impairment losses is symmetrical to the recognition of impairment losses in that the reversal will be recognized in the investment in associate account and not allocated to any component within the investment account. The investor compares the investment's recoverable amount (higher of value in use and fair value less costs to sell) with its carrying amount, whenever there is an indication that the investment may be impaired. Hence, impairment losses may be attributable to one of the three components: book value of net assets, fair value adjustments or goodwill.

For example, the past impairment of an investment in an associate was $250,000 and the current impairment is $100,000. The net profit before tax of the associate is $10,000,000 and the tax expense is $2,100,000. Ownership interest in the associate is 20%. The equity accounting entries are as follows:

Dr Opening retained earnings	250,000	
Cr Investment in associate		250,000

Past impairment loss

Explanatory note:

Since the investment in associate account includes goodwill at acquisition date, impairment losses are credited to the investment in associate account to write down the carrying amount of the investment account. This entry re-enacts past impairment losses. The impairment loss relates to the goodwill owned by the investor. Hence, there is no need to apply the ownership percentage to the impairment loss.

Dr Investment in associate	1,480,000	
Cr Share of profit of associate		1,480,000

Share of current profit after tax, after adjustment for current impairment loss

Share of profit after tax of associate	$1,580,000	(20% × $7,900,000)
Less current impairment loss	(100,000)	
Adjusted net profit after tax	$1,480,000	

Impairment loss relating to goodwill is assumed to be non-tax deductible

Explanatory note:

As with the past impairment loss, the current impairment relates wholly to the investor's share of the goodwill in the associate. There are no tax effects for reasons explained in Chapter 3.

As discussed above, for the purpose of goodwill impairment assessment, the recoverable amount of the investment in associate is the higher of the value-in-use and fair value less costs to sell under IAS 36. In determining the value-in-use of the investment, paragraph 42 of IAS 28 allows the investor to estimate either its share of the present value of the estimated future cash flows expected to be generated by the associate (which includes the cash flow from operations and the proceeds from the ultimate disposal of the investment) or the present value of the estimated future cash flows expected to arise from dividends to be received from the investment and its ultimate disposal. The IFRS Standard clarifies that using appropriate assumptions, both methods will provide the same results. The recoverable amount should be assessed for each associate individually unless the associate does not generate cash flows from continuing use that are largely independent on those from the other assets of the entity.

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It should also be emphasized that goodwill impairment testing for associates differs from the goodwill impairment testing for subsidiaries on a few aspects. In the case of goodwill relating to subsidiaries, paragraph 10(b) of IAS 36 requires goodwill acquired in a business combination to be tested for impairment annually. In contrast, IAS 28 requires an investor to test for impairment for the investment in associate, which includes the goodwill in accordance with the requirements of IAS 36 when there is objective evidence of impairment on occurrence of a loss event such as significant financial difficulty or probable bankruptcy of the associate or joint venture (IAS 28 Paragraphs 41A to 41C). Further, if an impairment loss is recognized, paragraph 104 of IAS 36 requires the impairment loss to be allocated first to reduce the carrying amount of any goodwill arising on business combination allocated to the cash-generating unit (or group of units) before allocating the rest to reduce the carrying amount of the other assets of the cash generating unit (or group of units) pro rata on the basis of the carrying amount of each asset in the cash generating unit (group of units). IAS 28, on the other hand, requires the investor to recognize the impairment loss against the entire carrying amount of the investment of associate without allocation to any asset including goodwill

that forms the carrying value of the investment. Finally, paragraph 124 of IAS 36 prohibits the reversal of any impairment loss recognized for goodwill relating to subsidiaries in subsequent periods. Conversely, as goodwill that forms part of the carrying amount of the investment in associate is not separately recognized and any impairment losses previously recognized reduces the carrying value of the investment in associate directly, IAS 28 and IAS 36 require the reversal of the impairment to be recognized to the extent of subsequent increases in the recoverable amount of the investment in associate.

When the Investor Applies the Equity Method in its Separate Financial Statements

If an investor uses the equity method in its separate financial statements, there is no need to re-enact equity accounting adjustments. Effectively, the balance of investment in associate in the investor's books would be identical to the balance of investment in associate in the consolidated financial statements or investors' financial statements prepared in place of consolidated financial statements (where the investor has associates but not subsidiaries).

When the Investor Carries the Investment at Fair Value in its Separate Financial Statements

If an investor uses the fair value method in its separate financial statements under IFRS 9, the investor needs to strip off the fair value changes from the investment in associate balance. It then proceeds to recognize the same equity accounting entries that are passed when the cost method is used. IFRS 9 requires the investor to measure the investment in equity securities at fair value through profit or loss (FVTPL). However, the investor is permitted to elect to measure the investments at fair value through other comprehensive income (FVOCI). IFRS 9 is discussed in greater depth in Chapter 9. If an investment in associate is measured at FVTPL, changes in fair value in the current period are taken to income statement. Past periods' changes in fair value would thus be included in the opening retained earnings of the investor. If the investor elects to measure the investments at FVOCI, changes in fair value in the current period are taken to the statement of profit or loss and other comprehensive income. Past periods' changes in fair value would be included in accumulated other comprehensive income (for example, fair value adjustment reserve in equity).

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Let's expand our earlier example as follows:

1. Initial investment in A as at 1 January 20x4 was $6,000,000.
2. Fair value as at the following reporting dates are as follows:

   31 December 20x4 $7,000,000

   31 December 20x5 $6,600,000

Assuming that the investor carries the investment at FVTPL, the entries passed in its books as a legal entity and the equity accounting entries passed for the economic entity would differ. Assume that the fair value changes have no tax consequences. Table 6.5 shows the accounting entries in the legal entity's books and the equity accounting adjustments.

TABLE 6.5 Investment in associate at fair value in separate financial statements

	Legal entity's books			Equity accounting adjustments		
1 January 20x4	Dr Investment	6,000,000		No adjustment		
	Cr Cash		6,000,000			
	Recognition of initial investment					
31 December 20x4	Dr Investment	1,000,000		Dr Fair value gain (I/S)	1,000,000	
	Cr Fair value gain (I/S)		1,000,000	Cr Investment		1,000,000
	Recognition of fair value gain			*Reversal of fair value gain*		
31 December 20x5	Dr Fair value loss (I/S)	400,000		Dr Investment	400,000	
	Cr Investment		400,000	Cr Fair value loss (I/S)		400,000
	Recognition of fair value loss			*Reversal of fair value loss*		
				Dr Opening retained earnings	1,000,000	
				Cr Investment		1,000,000
				Reversal of past period's fair value gain		

By stripping off the investment from the fair value changes, the equity accounting adjustments would bring the investment back to its original cost of $6,000,000. From this point, the usual equity accounting entries that apply to the investment carried at cost will apply to investments in associates carried at fair value. If the investment is carried at FVOCI, the adjustments would remove fair value changes taken to other comprehensive income.

Transfer of Assets between Investor and Associate

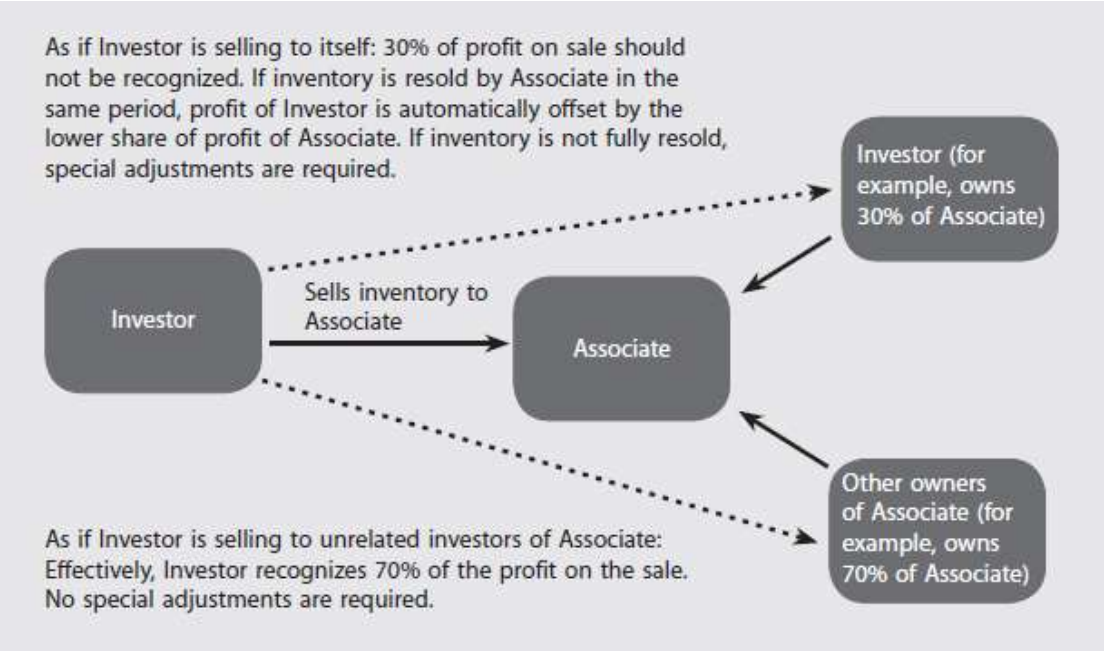
An investor's share in unrealized profit or loss from the transfer of assets to and from an associate must not be recognized in the share of profit from the associate. IAS 28 paragraph 28 states that:

> Gains and losses resulting from "upstream" and "downstream" transactions between an entity (including its consolidated subsidiaries) and its associate or joint venture are recognized in the entity's financial statements only to the extent of unrelated investors' interests in the associate or joint venture. "Upstream" transactions are, for example, sales of assets from an associate or a joint venture to the investor. "Downstream" transactions are, for example, sales or contributions of assets from the investor to its associate or a joint venture. The investor's share in the associate's or joint venture's gains and losses resulting from these transactions is eliminated.

In the equity method, the quantitative impact of adjustment for upstream and downstream sales is the same, unlike in consolidation where special considerations apply in the calculation of non-controlling interests' share of profit or loss. Non-controlling interests do not feature in the equity method as it requires the investor to recognize only its own share of profit and share of tax (which automatically excludes unrelated investors' share).

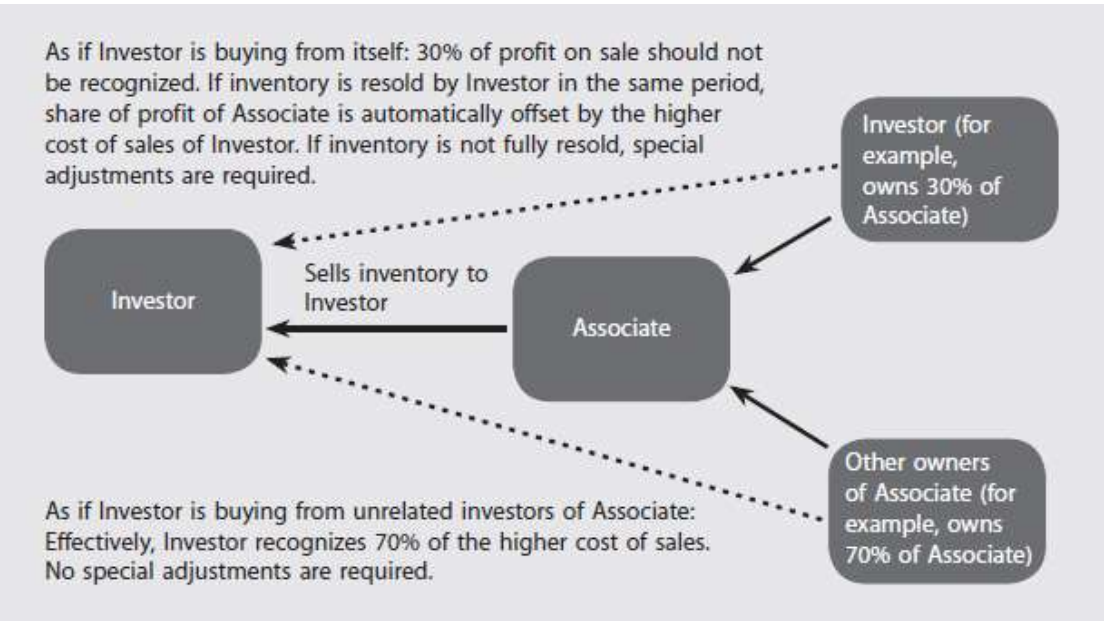
When an investor sells to its associate, it is as if the investor is selling partially to itself. Consider Figure 6.6, which illustrates the identity of the effective "purchaser" of an asset in a downstream transfer.

FIGURE 6.6 Downstream transfer



When an associate sells to its investor, it is as if the investor is buying partially from itself. Let us now consider the identity of the effective "seller" in an upstream transfer when an associate transfers an asset to its investor in Figure 6.7.

FIGURE 6.7 Upstream transfer



We can see from the diagrams that there is an "automatic" elimination of profit or cost of sales and share of profit of associates when the sale is realized through resale to third parties in the same period. However, when the sale is unrealized (that is, the asset remains in the books of either the investor or its associates), special adjustments must be effected to adjust the "unrealized profit." Special considerations for losses apply and are discussed later in the chapter.

IAS 28 does not prescribe the actual format for presenting the adjustments for unrealized profit on upstream and downstream transfers. Broadly speaking, there are two approaches to adjust the unrealized profit on a downstream or upstream transfer between an investor and an associate.

1. *Adjust the specific accounts affected.* For example, in an upstream sale, adjust equity accounted profit and the inventory of the investor; in a downstream sale, adjust the line items sales and cost of sales (COS) and investment in associate. When the profit is realized, adjust the specific line items that require correction. For example, in a subsequent year, when the transferred inventory is resold by the investor, the cost of sales of the investor should be adjusted together with equity accounted profit.
2. *Use the one-line adjustment approach to adjust the unrealized profit and the subsequent realization.* All adjustments are taken to the investment account and the share of profit for all transfers, whether upstream or downstream. Method 2 should be accompanied by footnote disclosures to explain the nature of the adjustments that affect the current share of profit of the associate.

In practice, there are many other variants that are a hybrid of Method 1 and Method 2. We apply Method 2 in our illustrations and problems, not only because it is simpler but it also better differentiates accounting for associates from accounting for subsidiaries. Transactions with associates are not eliminated by virtue of the fact that associates are not part of a "group." For example, if an investor makes a loan of $1,000,000 to its associate and earns interest income of $60,000 from the associate, the loan of $1,000,000 is not eliminated but left on the consolidated statement of financial position as a receivable. Likewise, the interest income is not eliminated on the consolidated income statement. However, by virtue of the offsetting effects through the share of profit, the investor effectively earns the unrelated investors' share of the interest income. If the ownership interest in the associate is 30% and if we ignore tax effects, the effects on the consolidated financial statements are as follows:

Consolidated income statement (extract)

Interest income	+ $60,000
Share of profit of associate	− $18,000
Net profit	+ $42,000

Consolidated statement of financial position (extract)

Investment in associate	−$18,000
Loan receivable from associate	+$1,000,000
Cash balances	−$1,000,000 + $60,000 = −$940,000
Net assets	$42,000
Retained earnings	+$42,000
Equity	+$42,000

This illustration shows that in a "downstream" transaction, the interest income is not eliminated, but shown in full. However, the associate recognizes the interest charged as an expense and consequently reports a profit that is lower by $60,000. The equity accounted profit will be lower by $18,000, that is, 30% × $60,000. The lower equity accounted profit partially offsets the interest income earned. The net effect on the consolidated retained earnings is to recognize up to 70% (unrelated investors' interests) of the interest income, that is, 70% × $60,000 or $42,000.

Similarly, the loan receivable from the associate is not eliminated on the consolidated statement of financial position. It is also wrong to proportionately eliminate the balance. Thus, there is a fundamental difference between the associate and the subsidiary. Effectively, the associate is akin to a "quasi-third party," whereby transactions and balances are not eliminated. However, the fundamental principle that IAS 28 upholds is that in substance, the investor should not recognize its own share of the profit on the transfer.

Method 2 is underpinned by the principle that the investor has a more distant relationship with the associate as compared with its relationship with the subsidiary. However, many practitioners may be uncomfortable with the distortions on the share of profit if a huge downstream adjustment is effected through the "one-line" equity accounted profit. In that respect, disclosure in the footnote is necessary to explain the components that are purely share of profit and reflective of the associate's performance and the components that are profit adjustments for transfers.

Consider the following example. An investor, I sells $200,000 worth of inventory to its associate A (A). The original cost of the inventory is $140,000. One-third of the inventory remains in A's warehouse at the end of the year. A's net profit before tax is $1,000,000, and its tax expense is $200,000. I has a 20% interest in A. Assume a tax rate of 20%.

A's net profit after tax	$800,000
Less after tax unrealized profit in the current year (80% × 1/3 × $60,000)	(16,000)
A's adjusted net profit after tax	$784,000
I's share of net profit at 20%	$156,800

Explanatory note:

The gross profit from the sale is $60,000 (sales price of $200,000 less original cost of $140,000). Since one-third of the inventory is unsold to third parties, I's share of the unrealized profit, that is, $4,000 (20% of 1/3 of $60,000), is not recognized. On an after-tax basis, this works out to be $3,200. Although I is not able to recognize its share of the unrealized profit, it recognizes 80% of the unrealized profit, that is, the portion that relates to the unrelated investor's share. Consider the overall effect on I's group financial statements:

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	I's profit (at group level) (Adjusted)	I's profit (at group level) (Unadjusted)
Gross profit from downstream sale	$ 60,000	$ 60,000
Share of A's profit	156,800	160,000
Profit effect	$216,800	$220,000

I is not able to recognize $3,200 or its share of the unrealized profit. However, I is able to recognize 80% of the unrelated investor's share as if it had sold the inventory to unrelated investors of A.

The equity accounting entry is as follows:

Dr Investment in A	156,800	
Cr Share of profit of A		156,800

In the case of downstream sales, the unrealized profit resides in the retained earnings of the investor's books. However, from the economic entity's perspective, the investor's share of the unrealized profit is unearned. To eliminate the investor's share of the unrealized profit, one approach is to reduce the investor's share of the unrealized profit as shown in the illustration above. Although the investor's share of the unrealized profit does not reside in the associate's books, the adjustment is effected through the share of profit of the associate in line with the concept of equity accounting, as a compressed single-line consolidation reflecting all relevant adjustments.[8] Hence, unrealized

profit from both upstream (associate's sale to investor) and downstream[9] (investor's sale to associate) sales is adjusted from the associate's net profit.

Let us now consider the impact of the adjustment on unrealized profit on an upstream transfer. We adapt the situation in the previous example such that associate A sells inventory to investor I. All other information remains the same. The investor is the buyer and will recognize a higher cost of sales as a result of the profit element in the transfer. Since two third of the inventory is sold, the investor will recognize an increase in cost of sales of $40,000 (that is, 2/3 × $60,000). Share of A's profit will be reduced by the share of the unrealized profit. You will note that in substance, both downstream and upstream sales have the same profit reduction of $3,200 to group profits.

	I's profit (at group level) (Adjusted)	I's profit (at group level) (Unadjusted)
Increase in cost of sales from upstream sales . . .	($40,000)	($40,000)
Share of A's profit.	156,800	160,000
Profit effect.	$116,800	$120,000

The difference between the adjusted and unadjusted amount is $3,200, which is I's share of the unrealized profit on the upstream transfer

In this example, both upstream and downstream adjustments are effected through the share of profit and investment in associate accounts. We are using the one-line adjustment approach in Method 2. In subsequent periods when the remaining one third of the inventory is resold to third parties, the same principle applies whereby adjustments are made through the share of profit line.

If Method 1 is applied, the specific line items will be adjusted. For example, in the upstream sales adjustment, the share of profit of A will be reduced together with a concurrent adjustment to the inventory of I. When the inventory is resold to third parties in subsequent periods, the cost of sales of I will be adjusted downwards.

More complex transfers of assets that constitute a business between an investor and an associate are dealt with in Appendix 6A.

Treatment of Dividends

Dividend income and other distributions received are deemed as repayment of profits and credited to the investment in associate account. Income is recognized purely on the basis of net profit of the associate. However, the separate financial statements of the investor measures investment in associate on the cost basis or as a financial asset under IFRS 9, whereby dividend is recognized as an income item. Hence, dividend income has to be reclassified from the income statement (I/S) and taken to the statement of financial position to reduce the investment in associate account. If the investor uses the equity method to measure the investment in associate in the investor's separate financial statements, the dividend would be shown as a reduction of investment. There is no need to make any equity accounting adjustments as the legal entity's investment balance would be identical with the balance required for the economic entity reporting under IFRS 10 (where consolidated financial statements are prepared) or IAS 28 (where the investor has only associates).

For example, dividend income from an associate is $160,000. The equity accounting adjustment if the investment is carried at cost or fair value is as follows:

Dr Dividend income (I/S)	160,000	
Cr Investment in associate		160,000

The above entry is a reclassification and not an elimination entry. Essentially, income from asso-ciates can be recognized only in one form, either as share of profits or as dividends, but not both. Hence, dividend income is reclassified to the statement of financial position as a realization of the accrued profits.

If the investor opts to use the equity method in its separate financial statements, the investor would pass the following entry on receipt of dividend income in its books:

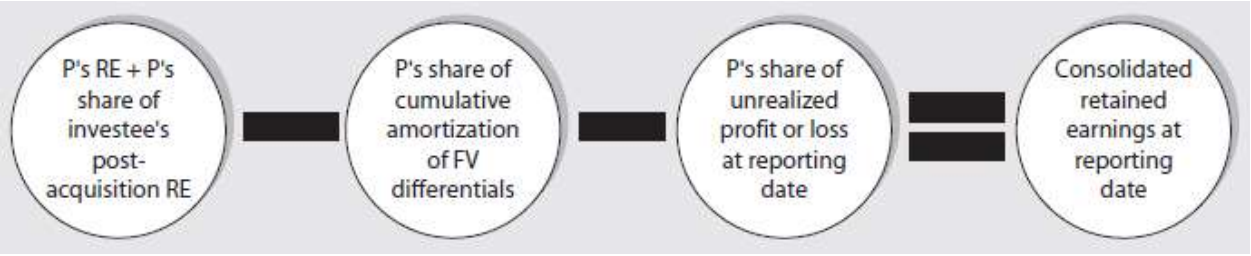
Dr Cash or Dividend receivable	160,000	
Cr Investment in associate		160,000

No further adjustment is required on consolidation if the investor uses the equity method in its separate financial statements. Unlike consolidation, equity accounting does not involve a line-by-line summation of an associate's financial statements. Hence, the dividends declared by the associate need not be eliminated.

PERFORMING AN ANALYTICAL CHECK OF CONSOLIDATED RETAINED EARNINGS WITH SUBSIDIARIES AND ASSOCIATES

Recall in Chapter 5 that we could determine the consolidated retained earnings analytically by considering the following three components. The analytical process can easily incorporate the impact of equity accounting on consolidated retained earnings. We reproduce Figure 5.3 here to show the three components.

FIGURE 6.8 Components of Consolidated Retained Earnings



When an investor has an associate, the analytical process is the same as for subsidiaries, except that there is no distinction for unrealized profit from upstream or downstream transfers. In accounting for subsidiaries, the parent would adjust for 100% of unrealized profits or losses from downstream transfers included in retained earnings but only its own share of unrealized profits or losses from upstream transfers. Non-controlling interests would have a share of adjustments from upstream transfers as well. However, in accounting for associates, the investor's interest is applied consistently to unrealized profits or losses from upstream or downstream transfers.

Illustration 6.1 explains the equity method in a specific example.

ILLUSTRATION 6.1 Equity accounting

P Co acquired a 40% interest in the ordinary shares of A Co on 1 January 20x2. P Co was deemed to have significant influence over A Co. P Co carried investment in A Co at cost in its separate financial statements. However, the

requirements of IAS 28 had to be complied with in the preparation of the financial statements of the economic entity where P Co's interests in the net assets of A Co were to be shown. The following information pertains to A Co at the date of acquisition and in the year ended 31 December 20x3.

A Co
Statement of Financial Position
As at 1 January 20x2

	A Co Book value	A Co Fair value
Fixed assets	$250,000	$300,000
Inventory	150,000	150,000
Accounts receivable	120,000	120,000
Cash	45,000	45,000
Total assets	$565,000	$615,000
Accounts payable	$245,000	$245,000
Share capital	200,000	
Retained earnings	120,000	
Equity and liabilities	$565,000	$615,000

A Co
Income Statement
For the Year Ended 31 December 20x3

	P Co	A Co
Sales	$8,000,000	$1,500,000
Cost of sales	(5,500,000)	(900,000)
Gross profit	$2,500,000	$ 600,000
Operating expenses net of other income	(1,000,000)	(400,000)
Net profit before tax	$1,500,000	$ 200,000
Tax	(300,000)	(40,000)
Net profit after tax	$1,200,000	$ 160,000
Dividends declared	(200,000)	(100,000)
Net profit attributable to shareholders	$1,000,000	$ 60,000
Retained earnings, 1 January 20x3	2,500,000	180,000
Retained earnings, 31 December 20x3	$3,500,000	$ 240,000

A Co
Statement of Financial Position
As at 31 December 20x3

	P Co	A Co

Investment in A Co	$ 600,000	
Fixed assets	4,000,000	$400,000
Inventory	2,000,000	200,000
Accounts receivable	1,000,000	35,000
Cash	120,000	20,000
	$7,720,000	$655,000
Accounts payable	$1,220,000	$215,000
Share capital	3,000,000	200,000
Retained earnings	3,500,000	240,000
	$7,720,000	$655,000

Additional information:

(a) Remaining useful life of fixed assets at the date of acquisition was four years with zero residual value.

(b) The following sales of inventory were made during 20x2 and 20x3:

	20x2	20x3
Sales from A to P	$300,000	
Original cost	(240,000)	
Gross profit on transfer	$ 60,000	
Percentage unsold to third parties at year-end	60%	10%
Sales from P to A		$600,000
Original cost		(450,000)
Gross profit on transfer		$150,000
Percentage unsold to third parties at year-end		30%

(c) A Co transferred fixed assets to P Co on 1 July 20x2. Details are as follows:

Date of transfer	1 July 20x2
Original cost	$100,000
Accumulated depreciation	(20,000)
Net book value at transfer date	$ 80,000
Transfer price	$120,000
Remaining useful life at date of transfer	4 years
Residual value is zero	

(d) There were no changes in the share capital of A Co from the date of acquisition.

(e) Assume a tax rate of 20%. Recognize tax on fair value adjustments.

Required:

1. Prepare the equity accounting (EA) entries for the year ended 31 December 20x3.
2. Perform an analytical check on the Investment in A Co as at 31 December 20x3.
3. Show P Co's financial statements for the year ended 31 December 20x3 after incorporating the effects of equity accounting of A Co.

1. Equity accounting (EA) entries for the year ended 31 December 20x3

EA1: Recognize share of post-acquisition retained earnings of A

Dr Investment in A	24,000	
Cr Opening retained earnings .		24,000

Retained earnings of A as at 1 January 20x3	$180,000	
Retained earnings of A as at date of acquisition	(120,000)	
Change in retained earnings	$ 60,000	
Share of A's change in retained earnings	$ 24,000	(40% × $60,000)

Explanatory note:

This entry re-enacts the cumulative effects of past equity accounting entries. The change in retained earnings from the date of acquisition to the beginning of the current period captures *cumulative post-acquisition profit* after taking into account dividends declared during the post-acquisition period. Since the separate financial statements carries the investment at cost, this entry is necessary to reinstate the equity accounted profits up to the end of the previous year.

EA2: Adjustment for unrealized profit in beginning fixed assets

Dr Opening retained earnings	12,800	
Cr Investment in A		12,800

Profit arising from transfer of fixed assets	$40,000	($120,000 – $80,000)
Tax on unrealized profit	(8,000)	
Unrealized after-tax profit . . .	$32,000	
P's share of unrealized after-tax profit	$12,800	(40% × $32,000)

Explanatory note:

This entry is necessary to remove the share of unrealized profit at the beginning of the reporting period, so that the opening retained earnings of the combined entity is equal to the closing retained earnings from the previous period. Note that the adjustment is only for the investor's interest in the unrealized profit and is on an after-tax basis. Since equity accounting does not entail a line-by-line combination of specific assets and liabilities, the adjustment for the "overstated" fixed asset and the deferred tax asset are effected against investment in A. The investment account effectively represents P's share of the net assets of A. Since opening retained earnings reflect after-tax cumulative profits, the tax effects of adjustments to opening retained earnings can be netted off from the adjustments.

EA3: Adjustment for past after-tax depreciation on unrealized profit in fixed assets

Dr Investment in A	1,600		
Cr Opening retained earnings		1,600	
Past "over-depreciation" on overstated fixed assets		$5,000	(50% × $40,000/4)
Less tax on over-depreciation		(1,000)	
Correction of after-tax over-depreciation		$4,000	
P's share of correction of after-tax over-depreciation		$1,600	(40% × $4,000)

Explanatory note:

As with EA2, the correction of excessive depreciation is made against the investment in A. Opening retained earnings are credited with the adjustment as the depreciation charge was overstated. Past depreciation was for half a year from 1 July 20x2 to 31 December 20x2.

EA4: Adjustment for past after-tax depreciation on undervalued fixed assets

Dr Opening retained earnings	4,000		(80% × 40% × $50,000/4)
Cr Investment in A		4,000	

Explanatory note:

The initial investment in A reflects P's share of fair value of A's identifiable net assets and goodwill. The excess of the fair value over book value is implicitly included in the investment account. Since the excess relates to an identifiable asset, the excess has to be amortized as P had effectively "purchased" a share of fixed asset at fair value. Under equity accounting, there is no separate recognition of fixed assets and the investment in A reflects the net assets of A. Hence, the depreciation is credited to investment in A. Since the depreciation relates to past periods, the debit entry is made to opening retained earnings. The tax effect of past depreciation reduces the debit charge to opening retained earnings.

EA5: Adjustment for unrealized profit in beginning inventory

Dr Opening retained earnings	11,520		
Cr Investment in A		11,520	
Transfer price		$300,000	
Original cost		(240,000)	
Profit on transfer		$ 60,000	
Percentage unsold as at 31 December 20x2		60%	
Unrealized profit as at 31 December 20x2		$ 36,000	(60% × $60,000)
P's share of unrealized profit		14,400	(40% × $36,000)
After-tax effect		11,520	(80% × $14,400)

Explanatory note:

This entry re-enacts the adjustment for unrealized profit as at the end of the previous period.

EA6: Reclassify dividend income as a reduction of investment

Dr Dividend income	40,000		
Cr Investment in A		40,000	(40% × $100,000)

Explanatory note:

This entry is not an elimination entry but a "reclassification" entry. Under the cost method, P records the net dividend income in the separate income statement. (The same applies if P had adopted the fair value basis under IFRS 9.) However, from an equity accounting perspective, the investor's share of the associate's profit is recognized as income. Hence, dividends have to be removed from the income statement to avoid double counting of income and credited to the investment account as a realization of capitalized profits.

EA7: Recognize share of current profit after tax of A

Dr Investment in A	58,400	
Cr Share of profit of A		58,400

Explanatory note:

This entry is the income recognition entry under the equity method. Effectively, the investor is "capitalizing" profit by increasing the investment account when profits are made by the associate. Share of profit of A has to be adjusted for current depreciation on undervalued asset, the recognition of realized profit from previous year's upstream sales, and the removal of unrealized profit from the current year's downstream sales. All figures are on an after-tax basis. The share of profit of A is shown below:

Profit after tax of A	$160,000	
Less after-tax depreciation of undervalued fixed asset (fair value adjustment)	(10,000)	(80% × $50,000/4)
Add after-tax correction of current "over-depreciation" of fixed assets (upstream sale)	8,000	(80% × $40,000/4)
Add after-tax realization of profit in beginning inventory (upstream sale)	24,000	(80% × 50% × $60,000)
Less after-tax unrealized profit included in ending inventory (downstream sale)	(36,000)	(80% × 30% × $150,000)
Adjusted profit after tax of A	$146,000	
Share of adjusted profit of A	$ 58,400	(40% × $146,000)

2. Analytical check on Investment in A Co

First, determine the balance in the investment in A, under the equity method, by adding the effects of the equity accounting entries.

Investment in A, at cost	$600,000

EA1: Share of post-acquisition retained earnings	24,000
EA2: Adjustment for beginning profit (after-tax) on transferred fixed assets	(12,800)
EA3: Adjustment for past "over-depreciation" (after-tax) on fixed assets	1,600
EA4: Adjustment for past depreciation (after-tax) of undervalued fixed assets	(4,000)
EA5: Adjustment for unrealized profit in (after-tax) beginning inventory	(11,520)
EA6: Dividends received	(40,000)
EA7: Share of current profit after tax	58,400
Investment in A as at 31 December 20x3	$615,680

Next, analyze the components of the investment in A. After applying equity accounting, the investment in A should comprise the following:

(1) Share of book value of shareholders' equity (or net assets) of A, after adjusting for outstanding unrealized profit or loss at the end of the year;

(2) Undepreciated or unamortized balance of fair value adjustment of identifiable net assets of A at the end of the year. The fair value adjustments relate to the difference between the fair and book value of identifiable net assets of A at the date of acquisition by P; and

(3) Balance of unimpaired goodwill that is included in investment in A at the end of the year.

These three familiar components that feature in the analytical check of non-controlling interests conversely apply to the investment in associate carried on the investor's consolidated statement of financial position.

We now show the results of an analytical check on the components of the investment in A as at 31 December 20x3.

Book value of shareholders' equity of A	$440,000	
Adjustment for outstanding unrealized after-tax profit from fixed asset transfer (upstream sale)	(20,000)	[80% × ($40,000 – $5,000 – $10,000)]
Adjustment for outstanding unrealized after-tax profit from inventory transfer (upstream sale)	(4,800)	(80% × 10% × $60,000)
Adjustment for outstanding unrealized after-tax profit (downstream sale)	(36,000)	(80% × 30% × $150,000)
Adjusted shareholders' equity of A	$ 379,200	
P's share of A's adjusted book value of equity (1)	$ 151,680	(40% × $379,200)
P's share of after-tax undepreciated balance of fair value adjustment of fixed assets (2)	$ 8,000	[40% × 80% × ($50,000 – $12,500 – $12,500)]
Implicit goodwill in investment in A:		
Initial investment	$ 600,000	
Less: Share of after-tax fair value of identifiable net assets of A at acquisition date	(144,000)	[40% × ($370,000 – $10,000) (Note 1)]

Goodwill implicit in investment in A (3)	$ 456,000
P's interest in A as at 31 December 20x3 (1) + (2) + (3)	$ 615,680

Note 1: Share of fair value of identifiable net assets of A at acquisition date

Book value of net assets of A at acquisition date	$320,000
Excess of fair value over book value of fixed assets	50,000
Fair value of net assets of A at acquisition date	$370,000
Deferred tax liability on excess of fair value over book value of fixed assets (20% × $50,000)	(10,000)
After-tax fair value of identifiable net assets of A	$360,000

Explanatory note:

The excess of fair value over book value of the fixed assets of A gives rise to future taxable income in the form of higher output of goods and services. Since the fair value of fixed assets is recognized at acquisition date, the deferred tax liability on the excess of fair value over book value should also be recognized under paragraph 19 of IAS 12 *Income Taxes*. Essentially, under IAS 12, the recognition of assets gives rise to a need to recognize the future tax payable arising from the realization of the assets.

3. P Co's financial statements for the year ended 31 December 20x3, after incorporating the effects of equity accounting of A Co

In this example, P Co has an investment in an associate but has no subsidiaries. The worksheet is prepared using the same format as a consolidation worksheet. The total debits and credits posted to the income statement are carried over to retained earnings in the statement of financial position. Investment in the associate is not eliminated, but is adjusted for share of profit and other equity accounting entries.

Income Statement and Partial Statement of Changes in Equity For Year Ended 31 December 20x3				
	P Co	Dr	Cr	P Co (with equity accounting)
Sales	$8,000,000			$8,000,000
Cost of sales	(5,500,000)			(5,500,000)
Gross profit	$2,500,000			$2,500,000
Operating expenses net of other income	(1,000,000)	$40,000		(1,040,000)
Share of profit of associate			$58,400	58,400
Net profit before tax	$1,500,000			$1,518,400
Tax	(300,000)			(300,000)
Net profit after tax	$1,200,000			$1,218,400
Dividends declared	(200,000)			(200,000)
Net profit attributable to shareholders	$1,000,000			$1,018,400
Retained earnings, 1 January 20x3	2,500,000	12,800	24,000	2,497,280
		4,000	1,600	
		11,520		
Retained earnings, 31 December 20x3	$3,500,000	$68,320	$84,000	$3,515,680

Statement of Financial Position As at 31 December 20x3				
	P Co	Dr	Cr	P Co (with equity accounting)
Investment in A Co	$ 600,000	$ 24,000	$ 12,800	$ 615,680
		1,600	4,000	
		58,400	11,520	
			40,000	
Fixed assets	4,000,000			4,000,000
Inventory	2,000,000			2,000,000
Accounts receivable	1,000,000			1,000,000
Cash	120,000			120,000
	$7,720,000	$ 84,000	$ 68,320	$7,735,680
Accounts payable	$1,220,000			$1,220,000
Share capital	3,000,000			3,000,000
Retained earnings	3,500,000	$ 68,320	$ 84,000	3,515,680
	$7,720,000	$ 68,320	$ 84,000	$7,735,680
Total adjustments		$152,320	$152,320	

Illustrations 6.2 and 6.3 are comprehensive problem sets incorporating consolidation and equity accounting.

ILLUSTRATION 6.2 Comprehensive problem

The income statements and summarized statements of financial position for P, S and A are presented below.

Income Statements
For Year Ended 31 December 20x4

	P	S	A
Operating profit	$1,375,000	$185,000	$365,625
Dividend income	77,500	0	0
Profit before tax	$1,452,500	$185,000	$365,625
Taxation	(275,000)	(37,000)	(73,125)
Profit after tax	$1,177,500	$148,000	$292,500
Dividends declared	(375,000)	(50,000)	(93,750)
Profit retained for the year	$ 802,500	$ 98,000	$198,750
Retained earnings, 1 January 20x4	1,806,875	652,000	363,750
Retained earnings, 31 December 20x4	$2,609,375	$750,000	$562,500

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Statements of Financial Position
As at 31 December 20x4

	P	S	A
Investment in S, at cost	$1,250,000		
Investment in A, at cost	875,000		
Other net assets	2,984,375	$1,625,000	$1,875,000
Net assets	$5,109,375	$1,625,000	$1,875,000
Share capital	$1,875,000	$ 625,000	$ 937,500
Other reserves	625,000	250,000	375,000
Retained earnings	2,609,375	750,000	562,500
Shareholders' equity	$5,109,375	$1,625,000	$1,875,000

Additional information:

(a)

	S	A
Date of initial investment by P	1 July 20x1	1 January 20x3
Number of shares purchased by P	500,000	375,000
Number of total issued shares	625,000	937,500

Retained earnings at the date of purchase by P	$450,000	$375,000

There are no changes in the share capital and other reserves since the date of initial investment. The fair value of non-controlling interests of S at acquisition date is $312,500.

(b) At the date of acquisition, S owned (and still owns) certain unrecognized identifiable intangible assets. All other assets acquired were at fair values.

Fair value of unrecognized licenses and trademarks at acquisition date	$50,000
Impairment in fair value during the year ended 31 December 20x3	$18,750

Book values of identifiable net assets of A are close to their fair values at the date of acquisition.

(c) In 20x3, the group recognized a goodwill impairment loss in S of $79,000. No goodwill impairment loss arose with respect to the investment in A.

(d) Information on intercompany transactions between P and S is as follows:

Year	Produced by	Sold to	Cumulative percentage resold to third parties		Original cost	Sales price
			20x3	20x4		
20x3	S	P	40%	90%	$37,500	$50,000
20x4	P	S		70%	$62,500	$93,750

Tax on the profit from these sales had been recognized in the individual books.

(e) Tax rate was 20%. Tax effects of fair value adjustments should be recognized in the consolidated financial statements.

Required:

1. Assuming that P had "control" over S and "significant influence" over A, show the consolidation and equity accounting entries, and their related tax effects that must be passed by P in the 20x4 group financial statements.
2. Show the non-controlling interests' balance as at 31 December 20x4. Perform a reconciliation check on the balance in non-controlling interests.
3. Show the balance of investment in A as at 31 December 20x4. Perform an analytical check on the balance of investment in A.

1. Consolidation adjustments

CJE1: Elimination of investment in S

Dr Share capital	625,000	
Dr Other reserves	250,000	
Dr Retained earnings, 1 January	450,000	
Dr Intangible asset	50,000	

Dr Goodwill	197,500	
Cr Investment in S		1,250,000
Cr Deferred tax liability		10,000
Cr Non-controlling interests		312,500

Explanatory notes:

- Investment in S was eliminated against the shareholders' equity of S, which resulted in the recognition of the intangible asset and goodwill.
- $$\text{Goodwill} = \underset{\text{transferred}}{\text{Consideration}} + \underset{\text{non-controlling interests}}{\text{Fair value of}} - \underset{\text{identifiable net assets of S}}{\text{Fair value of}}$$
  $$= \$1{,}250{,}000 + \$312{,}500 - \$1{,}365{,}000$$
  $$= \$1{,}562{,}500 - \$1{,}365{,}000$$
  $$= \$197{,}500$$
- Goodwill attributable to:
  $$\text{P} = \$158{,}000\ (\$1{,}250{,}000 - 80\% \times \$1{,}365{,}000)$$
  $$\text{Non-controlling interests} = \$39{,}500\ (\$312{,}500 - 20\% \times \$1{,}365{,}000)$$
- Fair value of identifiable net assets at acquisition date
  = Book value of identifiable net assets of S + Fair value of unrecognized intangible asset
  – Deferred tax liability on intangible asset
  = (Share capital + Other reserves + Retained earnings of S)
  + Fair value of unrecognized intangible asset – Deferred tax liability on intangible asset
  = ($625,000 + $250,000 + $450,000) + $50,000 – $10,000
  = $1,325,000 + $40,000
  = $1,365,000
- The book value of identifiable net assets of S at acquisition date is the book value of equity as reported in S's legal entity financial statements. Components of equity at acquisition date comprise share capital and "pre-acquisition" reserves and retained earnings.
- As an intangible asset of $50,000 was recognized at acquisition date, a corresponding deferred tax liability of $10,000 should be recognized on the asset under IAS 12. The recognition of the intangible asset implies that a future taxable income of $50,000 is to be expected from the intangible asset. The future taxable income gives rise to future tax payable of $10,000. The future tax payable is called deferred tax liability. The recognition of the asset warrants the recognition of the deferred tax liability that arises from the asset. page 360
- Non-controlling interests were recognized at fair value as at date of acquisition.
  $$\text{Non-controlling interests \%} = \frac{\text{Number of shares issued to non-controlling shareholders}}{\text{Total number of shares issued}}$$
  $$= \frac{125{,}000}{625{,}000} = 20\%$$

CJE2: Impairment of goodwill in 20x3

Dr Retained earnings, 1 January	63,200	
Dr Non-controlling interests	15,800	
Cr Accumulated impairment		79,000

Explanatory note:

Impairment of goodwill was recorded in a prior period in the consolidated financial statements. This group expense must be re-enacted in each subsequent year to opening retained earnings in order to ensure consistency between the group's opening retained earnings and the previous year's closing retained earnings of the group. Impairment of goodwill has no tax effects (IAS 12:15a). Non-controlling interests have a share of goodwill impairment.[10]

CJE3: Impairment of identifiable intangible asset in 20x3

Dr Retained earnings, 1 January	15,000	
Dr Non-controlling interests	3,750	
Cr Accumulated impairment		18,750

Explanatory note:

Since non-controlling interests were set up at fair value at acquisition date, this group of shareholders must also bear a share of the amortization effects of the fair value adjustments. In this case, the identifiable intangible asset was impaired in a prior period. Thus, non-controlling interests were debited with 20% of the impairment loss of $18,750. The group's share of the impairment was debited to opening retained earnings.

CJE4: Tax effects of impairment of intangible asset during 20x3

Dr Deferred tax liability	3,750	
Cr Retained earnings, 1 January		3,000
Cr Non-controlling interests		750

Explanatory note:

The impairment of the intangible asset results in the concurrent extinguishment of the deferred tax liability. Since the carrying amount of the intangible asset is reduced, the related deferred tax liability or future tax payable arising from the intangible asset is similarly reduced. As the impairment relates to a past period, the profit and loss effects are reflected in opening retained earnings. Furthermore, as non-controlling interests have a share of the deferred tax liability in CJE1, these parties would also have a share of the extinguishment of the deferred tax liability.

CJE5: Elimination of dividends from S

Dr Dividend income	40,000	
Dr Non-controlling interests	10,000	
Cr Dividends declared		50,000

Explanatory note:

Dividends declared of S are eliminated against the dividend income of P and reduction of non-controlling interests.

CJE6: Adjustment of unrealized profit in beginning inventory (upstream sale)

Dr Retained earnings, 1 January	6,000	
Dr Non-controlling interests	1,500	
Cr Cost of sales		6,250
Cr Inventory		1,250

Unrealized profit at beginning of year

Original sales price	$50,000
Original cost	(37,500)
Gross profit on sales from S to P	$12,500
% not yet sold to third parties at end 20x3	60%
% not yet sold to third parties at end 20x4	10%

Explanatory note:

Sixty percent of the inventory was unsold at the end of the previous year. Opening retained earnings were debited with the group's share of unrealized profit (80% × 60% × $12,500). Likewise, non-controlling interests were debited with their share of the adjustment of unrealized profit at the start of the year (20% × 60% × $12,500). The credit entries reflect the subsequent events relating to the inventory. Ten percent of the original inventory transferred still remained unsold at the end of the current year. Hence, inventory had to be credited (reduced) by 10% of the original profit (10% × $12,500). Fifty percent of the original inventory transferred were sold during the current year. Hence, the cost of sales had to be "brought down" to its original cost. The adjustment was 50% × $12,500, or $6,250, to remove the marked-up element from the cost of sales.

CJE7: Adjustment of tax on unrealized profit at start of year (upstream sale)

Dr Tax expense	1,250	
Dr Deferred tax asset	250	
Cr Retained earnings, 1 January		1,200
Cr Non-controlling interests		300

Explanatory note:

CJE7 reflects the tax effects of CJE6. As with all tax adjustments, the direction of the entries is opposite to the income adjustments.

CJE8: Elimination of intercompany sales and adjustment of unrealized profit in current year (downstream sale)

Dr Sales	93,750	
Cr Cost of sales		84,375
Cr Inventory		9,375

Explanatory note:

The elimination of downstream sales is a composite entry of two individual entries. To conceptually understand the entry, consider the two individual entries (a) and (b) below.

CJE8(a): Elimination of "realized sales"

Dr Sales (P's I/S)	65,625	
Cr Cost of Sales (S's I/S)		65,625

Explanatory note:

Seventy percent of the inventory purchased from P by S were resold to third parties by S during 20x4. Hence, 70% of the sales by P to S (70% × $93,750) were matched exactly to the cost of selling this inventory in S's books. The cost of sales was the amount that was invoiced by P. The profit and loss effects cancelled each other perfectly. An elimination entry is still required to remove intragroup transactions from the line items in the income statement.

CJE8(b): Elimination of "unrealized sales"

Dr Sales (P's I/S) 30% × $93,750	28,125	
Cr Cost of sales (P's I/S) 30% × $62,500 .		18,750
Cr Inventory (S's SFP)		9,375

Explanatory note:

Thirty percent of the inventory purchased from P by S still remained in S's inventory as at 31 December 20x4. The profit element that is included in the carrying amount of the inventory is deemed "unrealized" from the group's perspective. Hence, an adjustment is required to adjust out the unrealized profit from S's inventory in the statement of financial position and to reverse out the sales and cost of sales as recorded by P in its income statement.

CJE9: Adjustment of tax on unrealized profit in current year from downstream sale

Dr Deferred tax asset	1,875	
Cr Tax expense		1,875

Explanatory note:

The tax adjustment in CJE9 relates to the tax effects of CJE8(b). There are no tax effects in CJE8(a) as the income effects cancel out through the elimination entry.

CJE10: Allocation of undistributed post-acquisition retained earnings to non-controlling interests

Dr Retained earnings, 1 January	40,400	
Cr Non-controlling interests		40,400

Retained earnings at 1 January 20x4	$652,000
Retained earnings at acquisition date	450,000
Change	$202,000
Non-controlling interests' share at 20% . . .	$ 40,400

Explanatory note:

Non-controlling interests' share of post-acquisition retained earnings reduce the group's retained earnings.

CJE11: Allocation of share of current after-tax profit to non-controlling interests

Dr Income to non-controlling interests	30,600	
Cr Non-controlling interests		30,600

Profit after tax of S	$148,000
Add realized profit from prior-year transfer	6,250
Less tax on realized profit	(1,250)
Adjusted profit after tax	$153,000
Non-controlling interests' share at 20%	$ 30,600

Explanatory note:

The resale of transferred inventory to third parties warrants the recognition of profit to the economic entity. Since CJE6 reduces non-controlling interests for the unrealized profit at the start of the year, CJE11 is required to allocate to non-controlling interests the proportion that is realized during the year. The journal entry for equity accounting is shown below.

EA1: Equity-accounted profit for the year

Dr Investment in A	117,000	
Cr Share of A's profit after tax		117,000

Explanatory note:

A's profit after tax for 20x4 is $292,500. Since P has ownership interests of 40%, P recognizes 40% of $292,500 or $117,000 as share of profit. Applying the Illustrative Guidance in IAS 1 *Presentation of Financial Statements*, the profit is presented on an after-tax basis. However, companies may choose to present tax separately from income. P's ownership in A is 40% (375,000/937,500).

EA2: Dividend as a repayment of profits

Dr Dividend income	37,500	
Cr Investment in A		37,500

Explanatory note:

P's separate income statement recognizes dividend income of $37,500 from A. Under the equity method, income is recognized in the form of share of profit (EA1). As such, dividend income is deemed as a repayment of profit and reduces the amount of capitalized profit in the investment account.

EA3: Allocation of post-acquisition retained earnings to beginning of period

Dr Retained earnings, 1 January	4,500	
Cr Investment in A		4,500

Retained earnings at 1 January 20x4	$363,750
Retained earnings at acquisition date	(375,000)
Change	$(11,250)
P's share at 40%	$ (4,500)

Explanatory note:

EA3 recognizes P's share of change in post-acquisition retained earnings in opening retained earnings and investment in A. This entry is similar, but goes in the opposite direction to allocation of share of post-acquisition retained earnings to non-controlling interests. EA3 re-enacts the recognition of past equity accounted losses (in this example, retained earnings of A had declined after acquisition). If the associate had other comprehensive income (for example, revaluation reserves), the investor would also recognize share of change in post-acquisition other comprehensive income.

2. Non-controlling interests as at 31 December 20x4

Non-controlling interests as at acquisition date (CJE1)	$312,500
Share of past goodwill impairment (CJE2)	(15,800)
Share of prior-period impairment loss on intangible asset (CJE3)	(3,750)
Share of tax of prior-period impairment loss on intangible asset (CJE4)	750
Share of dividends declared (CJE5)	(10,000)
Share of adjustment of unrealized profit as at 1 January (CJE6)	(1,500)
Share of adjustment of tax on unrealized profit as at 1 January (CJE7)	300
Share of post-acquisition retained earnings as at 1 January (CJE10)	40,400
Share of current year's profit after tax (CJE11)	30,600
Non-controlling interests as at 31 December 20x4	$353,500

Reconciliation:

S's net assets as at 31 December 20x4	$1,625,000
Unrealized after-tax profit in inventory from upstream sale at 31 December 20x4 (10% × 80% × $12,500)	(1,000)
Unamortized after-tax balance of intangible asset 80% × ($50,000 – $18,750)	25,000
Adjusted S's net assets as at 31 December 20x4	$1,649,000
Non-controlling interests' share of adjusted S's net assets at 31 December 20x4	$329,800
Non-controlling interests' share of unimpaired goodwill ($39,500 – $15,800)	23,700
Non-controlling interests as at 31 December 20x4	$353,500

3. Analytical check on investment in associate balance as at 31 December 20x4

Initial investment			$875,000
EA1: Share of profit after tax			117,000
EA2: Dividends received			(37,500)
EA3: Share of post-acquisition retained earnings			(4,500)
Balance as at 31 December 20x4			$950,000
Analytical check:			
Share of book value of net assets of A as at 31 December 20x4			$750,000
Implicit goodwill included in investment in A:			
Cost of investment		$875.000	
Less share of fair value of net assets of A as at date of investment:			
Share capital	$ 937,500		
Other reserves	375,000		
Retained earnings	375,000		
Book value of net assets	$1,687,500		
Fair value adjustments	0		
Fair value of identifiable net assets	$1,687,500 × 40%	675,000	
Implicit goodwill included in investment in A			200,000
Investmet in A as at 31 December 20x4			$950,000

ILLUSTRATION 6.3 Comprehensive problem

P Ltd is an investor. As at 31 December 20x2, P Ltd held shares in three companies, as follows:

	SA Co	SB Co	A Co
Percentage shareholding held by P Ltd	80%	100%	30%
Date of acquisition	1 Jan 20x1	1 Jan 20x2	1 Jan 20x2
Purchase consideration paid by P Ltd for the shares	$5,000,000	$4,600,000	$1,800,000

Shareholders' equity of companies at date of acquisition:

Share capital	$2,000,000	$1,000,000	$3,000,000
Retained earnings	2,200,000	2,600,000	3,000,000
	$4,200,000	$3,600,000	$6,000,000

Fair value of non-controlling interests in SA Co as at date of acquisition	$1,250,000
Excess of fair value over book value of buildings of SB Co	$600,000

P Ltd has control of SA Co and SB Co but only significant influence over A Co. The book values of the identifiable net assets of SA Co and A Co were close to their fair values. Goodwill of SA Co attributable to both P Ltd

and non-controlling interests was impaired to the extent of $164,000 in 20x1.

The excess of purchase consideration of P Ltd's interest in shareholders' equity of the wholly owned SB Co at the acquisition date was attributable to an undervaluation of buildings of $600,000 with the remainder attributable to goodwill. Goodwill impairment was 10% of original goodwill in the first year of acquisition. The undervaluation of buildings was to be adjusted in the consolidated statement of financial position and depreciated on a straight line basis over a useful life of 20 years.

Sales of P Ltd during the year included intercompany sales to SA Co amounting to $300,000, which were priced at 25% above cost. The inventory of SA Co included unrealized profits of $30,000 as at 1 January 20x2 and $25,000 as at 31 December 20x2. Inventories were normally sold within one year of purchase. Tax rate was 20%. Deferred tax liability on fair value adjustments should be provided for.

The financial statements of P and the other companies for 20x2 are shown below.

Required:

1. Prepare the consolidation and equity accounting journal entries for the year ended 31 December 20x2.
2. Determine the following amounts and balances in the consolidated financial statements for 20x2:

   (a) Operating profit
   (b) Interest income
   (c) Interest expense
   (d) Fixed assets
   (e) Inventory
   (f) Non-controlling interests in SA Co
   (g) Investment in A Co
   (h) Retained earnings

Income Statement
For Year Ended 31 December 20x2

	P	SA Co	SB Co	A Co
Sales	$5,600,000	$3,500,000	$980,000	$2,500,000
Operating profit	$2,305,000	$ 666,250	$328,750	$ 876,000
Dividend income	280,000	0	0	0
Interest income from subsidiaries and associate	40,000	0	0	0
Interest expense to P	0	(20,000)	(15,000)	(5,000)
Interest expense – others	(77,000)	(60,000)	(25,000)	(3,000)
Profit before tax	$2,548,000	$ 586,250	$288,750	$ 868,000
Tax	(508,900)	(117,250)	(57,750)	(173,600)
Profit after tax	$2,039,100	$ 469,000	$231,000	$ 694,400

Extract from Statement of Changes in Equity (Retained Earnings)				
	P	SA Co	SB Co	A Co
Balance as at 1 January	$6,700,000	$4,000,000	$2,600,000	$3,000,000
Prior-period item		120,000		(40,000)
Profit for the year	2,039,100	469,000	231,000	694,400
Dividends	(500,000)	(210,000)	(70,000)	(140,000)
Balance as at 31 December	$8,239,100	$4,379,000	$2,761,000	$3,514,400

Statement of Financial Position As at 31 December 20x2				
	P	SA Co	SB Co	A Co
Share capital	$10,000,000	$2,000,000	$1,000,000	$3,000,000
Retained earnings	8,239,100	4,379,000	2,761,000	3,514,400
Shareholders' equity	$18,239,100	$6,379,000	$3,761,000	$6,514,400
Fixed assets	$5,900,000	$6,000,000	$3,500,000	$5,900,000
Investment in subsidiaries	9,600,000	0	0	0
Investment in associate	1,800,000	0	0	0
Loans to subsidiaries	850,000	0	0	0
Loan to associate	120,000	0	0	0
Current assets:				
Inventory	1,461,900	751,000	779,000	1,197,600
Accounts receivable	190,000	450,000	456,000	250,000
Dividends receivable from subsidiaries	70,000	0	0	0
Dividends receivable from associate	12,000	0	0	0
Cash and bank	39,100	1,000	5,000	10,000
	$ 1,773,000	$1,202,000	$1,240,000	$1,457,600
Current liabilities:				
Bank loans	565,000	105,750	171,250	389,600
Accounts payable	630,000	100,000	320,000	120,000
Dividends payable	100,000	50,000	30,000	40,000
Provision for taxes	308,900	67,250	27,750	93,600
	$ 1,603,900	$ 323,000	$ 549,000	$ 643,200
Net current assets	169,100	879,000	691,000	814,400
Deferred tax liabilities	(200,000)	(50,000)	(30,000)	(80,000)
Loans payable to P	0	(450,000)	(400,000)	(120,000)
Net assets	$18,239,100	$6,379,000	$3,761,000	$6,514,400

1. Consolidation and equity accounting entries

The journal entries to show elimination of investment in subsidiaries are shown below.

CJE1a: Elimination of investment and shareholders' equity in SA

Dr Share capital	2,000,000	
Dr Retained earnings	2,200,000	
Dr Goodwill	2,050,000	
Cr Investment in SA		5,000,000
Cr Non-controlling interests		1,250,000

$$\text{Goodwill of SA} = \left(\begin{array}{c}\text{Investment in SA by P}\\+\\\text{Fair value of non-controlling interests}\end{array}\right) - \begin{array}{c}\text{Fair value of identifiable net assets}\\\text{of SA at acquisition date}\end{array}$$

$$= (\$5{,}000{,}000 + \$1{,}250{,}000) - \$4{,}200{,}000$$
$$= \$6{,}250{,}000 - \$4{,}200{,}000$$
$$= \$2{,}050{,}000$$

CJE1b: Elimination of investment and shareholders' equity in SB

Dr Share capital	1,000,000	
Dr Retained earnings	2,600,000	
Dr Buildings	600,000	
Dr Goodwill	520,000	
Cr Investment in SB		4,600,000
Cr Deferred tax liability		120,000

$$\text{Goodwill of SB} = \begin{array}{c}\text{Investment}\\\text{in SB by P}\end{array} - \left(\begin{array}{c}\text{Fair value of identifiable net assets}\\\text{of SB at acquisition date}\end{array}\right)$$

$$= \$4{,}600{,}000 - (\$3{,}600{,}000 + \$600{,}000 - \$120{,}000)$$
$$= \$4{,}600{,}000 - \$4{,}080{,}000$$
$$= \$520{,}000$$

Next, we show the journal entries for impairment of goodwill and depreciation of undervalued building.

CJE2: Past cumulative impairment of goodwill in SA

Dr Opening retained earnings	131,200	
Dr Non-controlling interests	32,800	
Cr Accumulated impairment		164,000

CJE3: Current impairment of goodwill in SB

Dr Impairment of goodwill	52,000	
Cr Accumulated impairment		52,000

CJE4: Depreciation on undervalued buildings

Dr Depreciation	30,000	
Cr Accumulated depreciation		30,000

Depreciation on the undervaluation: $600,000/20 years

CJE5: Tax effects of depreciation on undervalued buildings

Dr Deferred tax liability	6,000	
Cr Tax expense		6,000

Tax on depreciation in CJE4: 20% × $30,000

The journal entries to show adjustment of unrealized profits from intragroup transactions are:

CJE6: Realization of profit in opening inventory (downstream sale)

Dr Opening retained earnings	30,000	
Cr Cost of sales		30,000

CJE7: Tax expense on realized profit in opening inventory (downstream sale)

Dr Tax expense	6,000	
Cr Opening retained earnings		6,000

CJE8: Unrealized profit in closing inventory (downstream sale)

Dr Sales	300,000	
Cr Cost of sales		275,000
Cr Inventory		25,000

CJE9: Tax expense on unrealized profit in closing inventory (downstream sale)

Dr Deferred tax asset	5,000	
Cr Tax expense		5,000

The journal entries for the elimination of intragroup transactions and balances are:

CJE10: Elimination of intragroup interest income and expense

Dr Interest income	35,000	
Cr Interest expense		35,000

CJE11: Elimination of intragroup loan balances

Dr Loans payable to P	850,000	

Cr Loans receivable from subsidiaries .		850,000

CJE12: Elimination of intragroup payables and receivables

Dr Dividends payable	70,000	
Cr Dividends receivable		70,000

CJE13: Elimination of dividend income and dividends declared

Dr Dividend income	238,000	
Dr Non-controlling interests	42,000	
Cr Dividends declared		280,000

The journal entries for non-controlling interests are:

CJE14: Allocation of post-acquisition profits to non-controlling interests

Dr Opening retained earnings	360,000	
Cr Non-controlling interests		360,000

Retained earnings at 1 January 20x2 .	$4,000,000
Retained earnings at acquisition date .	2,200,000
Change in retained earnings	$1,800,000
Non-controlling interests' share at 20%	$ 360,000

CJE15: Non-controlling interests' share of current profit after tax and prior-period items

Dr Income to non-controlling interests	93,800	
Dr Prior-period item (opening retained earnings)	24,000	
Cr Non-controlling interests		117,800

The non-controlling interests' deduction in the income statement relates to their share of profit for the year, excluding prior-period items. Prior-period items are adjustments to retained earnings for correction of errors or changes in accounting policies. Non-controlling interests' share of the prior-period items is debited directly to opening retained earnings because the prior-period deduction in the statement of retained earnings is a line item that is separate from profit after tax. Hence, non-controlling interests' share of prior-period item should not be included in the "normal" income allocation to non-controlling interests in the income statement.

Finally, we show the journal entries for equity accounting of associate:

EA1: Equity-accounted profits of associate

Dr Investment in associate	196,320	
Dr Share of associate's prior-period item* .	12,000	

Cr Share of associate's profit		208,320

* Adjustment to opening retained earnings

EA2: Reclassification of dividends as a repayment of profit

Dr Dividend income	42,000	
Cr Investment in associate		42,000

The investor recognizes a share of all changes in equity of the associate. In this illustration, the associate recognizes a prior-period adjustment to opening retained earnings. The investor equity accounts for a share of the prior-period adjustment. The same approach applies to other changes in equity such as other comprehensive income.

2. Consolidated financial statements

There are various ways to determine the amounts and balances as reported on the consolidated financial statements as at 31 December 20x2. Essentially, we can consider two ways. One is the straight-forward worksheet or listing approach, which is simply a compilation of the sum of the legal entity amounts and the consolidation adjustments. The other may be broadly described as the analytical approach, which derives the consolidated amounts and balances independently of the consolidation adjustments. However, when the calculations are less complex, the two approaches tend to be rather similar. However, in more complex items, such as non-controlling interests, the two approaches would be quite different and the analytical approach serves to be a more effective check on the worksheet approach.

(a) *Operating profit*

Worksheet approach:

P	$2,305,000
SA Co	666,250
SB Co	328,750
CJE3	(52,000)
CJE4	(30,000)
CJE6	30,000
CJE8	(25,000)
	$3,223,000

Analytical approach:

P	$2,305,000
SA Co	666,250
SB Co	328,750
Goodwill impairment in SB	(52,000)
Depreciation of undervalued fixed assets	(30,000)
Realized profit from 20x1 transfers	30,000
Unrealized profit from 20x2 transfers	(25,000)
	$3,223,000

(b) *Interest income from subsidiaries and associates*

Worksheet approach:

P	$40,000
SA Co	–
SB Co	–
CJE10	(35,000)
	$ 5,000

Analytical approach: Only interest income from third parties should be recognized.

Interest income from associate	$5,000

(c) *Interest expense*

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Worksheet approach:

P	$ 77,000
SA Co	80,000
SB Co	40,000
CJE10	(35,000)
	$162,000

Analytical approach: Only interest expense to third parties should be recognized.

P Co	$ 77,000
SA Co	60,000
SB Co	25,000
	$162,000

(d) *Fixed assets*

Worksheet approach:

P	$ 5,900,000
SA Co	6,000,000
SB Co	3,500,000
CJE1b	600,000
CJE4	(30,000)
	$15,970,000

Analytical approach:

P	$ 5,900,000
SA Co	6,000,000

SB Co	3,500,000
Add balance of undervalued building	570,000
	$15,970,000

(e) *Inventory*

Worksheet approach:

P	$1,461,900
SA Co	751,000
SB Co	779,000
CJE8	(25,000)
	$2,966,900

Analytical approach:

P	$1,461,900
SA Co	751,000
SB Co	779,000
Unrealized profit in inventory at 31 Dec 20x2	(25,000)
	$2,966,900

(f) *Non-controlling interests in SA Co*

Worksheet approach:

CJE1a	$1,250,000
CJE2	(32,800)
CJE13	(42,000)
CJE14	360,000
CJE15	117,800
	$1,653,000

Analytical approach:

Book value of equity of SA Co	$6,379,000
Non-controlling interests' share of adjusted book value of equity	$1,275,800
Non-controlling interests' goodwill	
Fair value of non-controlling interests at acquisition date	1,250,000
Less non-controlling interests' share of fair value of identifiable net assets as at acquisition date	(840,000)
Goodwill	$ 410,000

Less impairment loss	(32,800)
Unimpaired goodwill attributable to non-controlling interests	$ 377,200
Non-controlling interests' balance	$1,653,000

(g) *Investment in Associate*

Worksheet approach:

Investment in associate, at cost	$1,800,000
EA1	196,320
EA2	(42,000)
	$1,954,320

Analytical approach:

Book value of equity of A Co	$6,514,400
P's share of book value of equity of A Co	$1,954,320
Investment in associate, at cost	1,800,000
Less share of fair value of identifiable net assets at investment date	(1,800,000)
Goodwill	–
Investment in associate	$1,954,320

(h) *Retained earnings*

Worksheet approach:

P	8,239,100
SA Co	4,379,000
SB Co	2,761,000
CJE1a	(2,200,000)
CJE1b	(2,600,000)
CJE2	(131,200)
CJE3	(52,000)
CJE4	(30,000)
CJE5	6,000
CJE8	(25,000)
CJE9	5,000
CJE13	42,000
CJE14	(360,000)
CJE15	(117,800)
EA1	196,320

EA2	(42,000)
	10,070,420

Analytical approach:

P	8,239,100	
P's share of SA Co's post-acquisition retained earnings	1,743,200	80% × (4,379,000 − 2,200,000)
P's share of SB Co's post-acquisition retained earnings	161,000	100% × (2,761,000 − 2,600,000)
P's share of A Co's post-acquisition retained earnings	154,320	30% × (3,514,000 − 3,000,000)
Less:		
P's share of goodwill impairment of SA Co	(131,200)	80% × 164,000
P's share of goodwill impairment of SB Co	(52,000)	100% × 10% × 520,000
P's share of after-tax depreciation of undervalued buildings of SB Co	(24,000)	100% × 80% × 600,000/20
Less:		
P's share of unrealized profit, after tax, of downstream transfer	(20,000)	100% × 80% × 25,000
	10,070,420	

SPECIAL ISSUES RELATING TO THE EQUITY METHOD

Transfers at a Loss

When an investor transfers an asset to the associate at a loss ("downstream transfer"), the investor has to evaluate whether the loss is impairment in nature. The newly revised IAS 28 clarifies that the impairment loss to be recognized in full by the investor. In other words, impairment loss should be recognized in full before inventory is released. For example, an investor sells inventory at fair value of $30,000 to its associate in which it has a 30% interest. The carrying amount of the inventory is $50,000. The investor should recognize $20,000 as loss in the income statement. There will not be any difference in treatment for realized or unrealized sales. In this scenario, assume that the associate sells 90% at fair value of $27,000 and holds the remaining 10% in inventory at $3,000. The net profit earned by the associate is zero. For simplicity, we assume that there is no assurance that the tax loss could be used to offset future income, and hence, we do not recognize the tax benefit on the loss. The consolidated income statement will show the following income items:

Revenue	$30,000
Cost of sales	(50,000)
Gross loss	($20,000)
Share of profit of associate	–
Net loss	($20,000)

The loss has to be recognized in full by the investor. However, if the associate sells inventory to investor ("upstream transfer") at fair value of $30,000 when the carrying amount is $50,000, the investor recognizes its share of the loss rather than the full loss of $20,000. Let us assume that the investor sells 90% at fair value of $27,000 and holds the remaining 10% in inventory at $3,000. The investor recognizes $6,000 or 30% of $20,000 as the share of loss of the investor. The fact that 10% of the inventory is unsold is not important as the loss is indicative of impairment of the inventory. The consolidated income statement will show the following income items:

Revenue	$27,000
Cost of sales	(27,000)
Gross loss	–
Share of loss of associate	($6,000)
Net loss	($6,000)

Applying the same principles in Chapter 5, if the loss is an "artificial" loss, we should adjust it out in the same manner as we do for unrealized profit. We would add back the artificial loss that is unrealized at the end of the year. In subsequent years, we should deduct the realized loss on the portion that is sold. Recall that a loss is artificial if the asset is not impaired but is transferred at an artificially low price. For example, the inventory may be transferred at $30,000 when its carrying amount is $40,000 and the fair value is $50,000. The inventory is not impaired and the loss on sale of $10,000 is artificial. The various situations of impairment and artificial loss that were analyzed in Chapter 5 apply to identifying the nature of the loss in transfers between an investor and its associate.

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Can the Investment in an Associate be a Negative Balance?

When post-acquisition losses exceed the initial investment in an associate, the investor should stop equity accounting of further losses. In other words, the investment is carried at zero balance. An exception to the requirement arises when the investor has a constructive or legal obligation to make good the losses (IAS 28:39). In this matter, the accounting for investments in associates is different from accounting for non-controlling interests. Recall, in Chapter 5, non-controlling interests become a deficit balance when the post-acquisition losses exceed the initial amount of the non-controlling interests. However, the same does not apply to equity accounting.

Other Issues

The investor should also compare the reporting dates and accounting policies of its associates. If the reporting dates and/or accounting policies are not the same as the investor's, the investor has to make appropriate adjustments to the financial information of the associate before applying the equity method. Ultimately, the financial information of the associate must conform to the reporting date and accounting policies of the investor.

Occasionally, an associate may have outstanding cumulative preference shares that are held by other parties and these are classified as equity. In computing the share of profit of the associate, the investor should deduct the dividends on these shares from the profit, irrespective of whether the dividends have been declared or not (IAS 28:37).

ACCOUNTING FOR JOINT ARRANGEMENTS

This chapter thus far has explained and illustrated the principles and processes of the equity method. We also saw how the equity method applies to the accounting for investments in associates in the economic entity financial statements. An associate is an investee over which the investor has significant influence.

We now consider another type of investment in which the equity method applies. These investments are described as joint arrangements. A joint arrangement exists when two or more parties to the arrangement has joint control. In the introduction to this chapter, we explained what joint control is. Essentially, joint control is the contractual sharing of power among parties such that decisions about relevant activities require the unanimous consent of these parties. Hence, a joint arrangement has two characteristics: (a) the existence of a contractual arrangement and (b) the parties to the contract has joint control over the arrangement.

In this section, we go deeper into the nature of a joint arrangement and the accounting methods that apply. One of the methods, the equity method, is already explained in the earlier part of this chapter. IAS 28 not only deals with accounting for associates but also accounting for joint ventures. However, we will also discuss another accounting method that applies to joint operations.

IFRS 11 *Joint Arrangements* is the standard that governs the accounting for joint arrangements. IFRS 11 applies to financial periods beginning on or after 1 January 2013 and establishes accounting principles for financial reporting by parties to a joint arrangement. IFRS 11 supersedes IAS 31 *Interests in Joint Ventures* and SIC 13 *Jointly Controlled Entities — Non-Monetary Contributions by Venturers*.

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By replacing IAS 31, IFRS 11 seeks to improve the accounting for joint arrangements in two ways. First, IFRS 11 removes the choice of accounting methods that IAS 31 permits. In the past, joint venture parties in a jointly controlled entity could choose either the proportionate consolidation or the equity method. In practice, the equity method was more popular because it reveals less and has a more favorable impact on leverage. Proportionate consolidation was the consolidation of pro-rated line items on the income statement and statement of financial position of the jointly controlled entity. Proportionate consolidation would increase the assets and liabilities of the joint venturer. As a consequence, ratios such as return on assets, debt to equity ratio and leverage ratios would be more negatively impacted.

Joint Ventures and Joint Operations

There are essentially two types of joint arrangements: joint operations and joint ventures. IFRS 11 defines the joint operation as a "joint arrangement whereby the parties that have joint control of the arrangement have rights to the assets, and obligations for the liabilities, relating to the arrangements. Those parties are called joint operators."[11] For example, a joint operator may contribute knowledge and expertise while another operator may contribute equipment. The first operator would have rights to the intellectual property of the joint arrangement while the second operator would have rights to property, plant, and equipment. The operators may also have different obligations depending on the contractual arrangements.

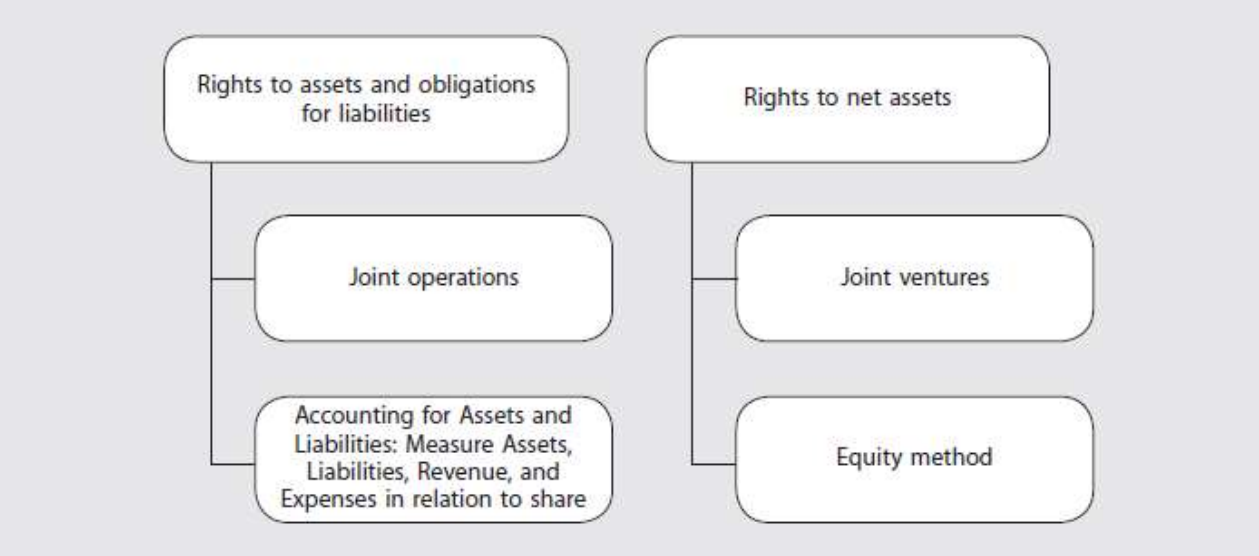
Joint operations need not be in a form of a separate vehicle such as a third entity. Project accounts are maintained for each joint operation. Each joint operator will have a share of revenues or share of output from the joint operation and bear own expenses and share of expenses in proportions that are agreed upon in the contractual arrangement. Each operator will also recognize own assets and own liabilities and share of assets and share of liabilities from the joint operation as specified in the agreement. Joint operators may provide guarantees in their own names to obtain loan financing. Each operator will contribute assets and/or expertise to the project and bear liabilities according to the contractual agreement.

Joint operations are fairly common in mining and extractive industries. Projects in these industries are highly capital intensive with substantial need for equipment, expertise, and financing. Joint operations provide resources from different joint operators to better manage the scale of operations and the risks that result from these projects.

The other joint arrangement is a joint venture. IFRS 11 defines a joint venture as an arrangement "whereby the parties that have joint control of the arrangement have rights to the net assets of the arrangement. Those parties are called joint venturers." Unlike a joint operation, a joint venture does not give rise to rights to specific assets and obligations for specific liabilities. All parties to a joint venture would have rights to the net assets or equity in the arrangement. For example, if two parties have a contractual arrangement to set up a company to develop a new drug, both parties will have rights to the net assets of the company. In many ways, this arrangement is similar to the relationship that an investor has with its associate. However, the difference between significant influence and joint control is the existence of a contractual arrangement in joint control. The sharing of power has to be determined by a contract.

It is important to identify the type of joint arrangement as the accounting methods differ for each. Figure 6.9 summarizes the approaches in IFRS 11.

FIGURE 6.9 Types of joint arrangements

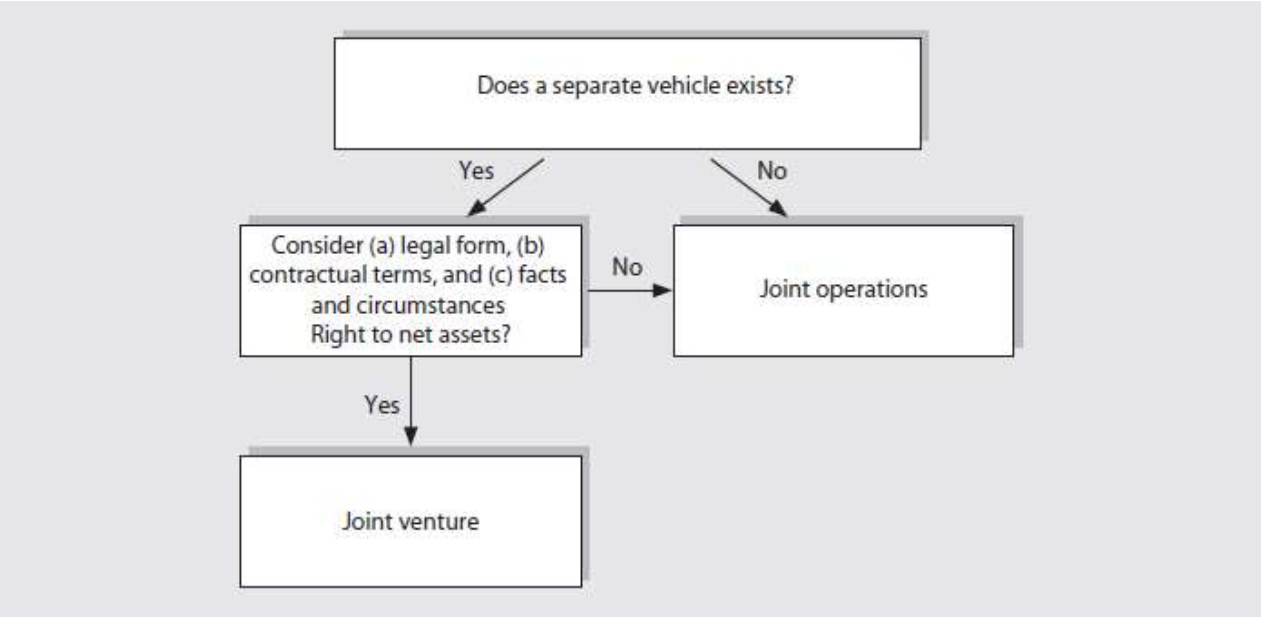


The determination of the type of joint arrangement often involves judgement. Parties to a joint arrangement must consider the facts and circumstances to determine whether the joint arrangement is effectively a joint operation or a joint venture. Very often, a joint arrangement is structured through a separate vehicle. The form of the vehicle may differ but it may well be a separate company. "A joint arrangement that is not structured through a separate vehicle is a joint operation."[12] In a joint operation, a contract determines each joint operator's rights to the assets and obligations for the liabilities of the joint arrangement, and each operator's rights to the revenues and obligations for expenses of the arrangement. In a joint venture, each joint venturer (owner) has rights to net assets of a joint arrangement (entity). Each joint venturer does not have rights to the assets or obligations to the liabilities of the joint arrangement. The separation of entity from its owners is a distinguishing feature in a joint venture. Hence, a joint venture requires the existence of a separate vehicle whose legal form confers it with rights and obligations that are separate from those of its owners (the joint venturers). However, IFRS 11 also explains that assets and liabilities held through a separate vehicle may be either a joint venture or a joint operation.[13]

The acid test is still with respect to whether each joint arrangement party has rights to net assets or rights to assets and obligations of the joint arrangement. In a joint operation, a separate vehicle, if any, would be of a legal form that allows the owners (the joint operators) to have rights to assets and obligations for the liabilities of the separate vehicle. The separate vehicle in a joint arrangement would typically not be an incorporated company or any other structure that is recognized as a legal person with rights and obligations that are separate from its owners.

In a nutshell, a joint venture is structured through a separate vehicle. However, the presence of a separate vehicle does not imply that the joint arrangement is a joint venture. The acid test has to be applied to determine the rights and obligations of each party to the joint arrangement. Figure 6.10 shows the decision flow.

FIGURE 6.10 Separate vehicles in joint arrangements



In some cases, a joint arrangement has to be inferred from the facts and circumstances. Unanimous consent implies that no one party in a contractual arrangement is able to make decisions about relevant activities unilaterally. Recall in Chapter 2 that relevant activities are those activities that have a significant impact on returns. Determination of relevant activities is context specific and one must evaluate the significance of each activity in relation to the returns of the joint arrangement. As with control, joint control is not the power to exercise protective rights. The rights must be substantive, albeit shared with other counterparties in unanimous decision making. In Chapter 2, we explained that protective rights are exercisable only in rare circumstances as defined by the contractual agreement (for example, a default of payment).

Unanimous consent need not always entail a 100% agreement from all parties to the contract. A contractual arrangement often stipulates the minimum required proportion of voting rights[14] required to determine a decision on relevant activities of another entity. For example, a shareholder agreement among three investors stipulates that the minimum vote required to approve the appointment of directors is 75%. In this example, unanimous consent is achieved by 75% votes. The appointment of directors is clearly an important decision that affects how relevant activities of the entity are managed. Consider the following independent scenarios. The following table shows the percentage of ownership interests held by each investor.

	Scenario 1	Scenario 2	Scenario 3
Investor A	40%	50%	35%
Investor B	40%	25%	30%
Investor C	20%	25%	5%
Other investors (<1% each)			30%
Total	100%	100%	100%

Scenario 1: A and B collectively, but not individually, are able to decide on the appointment of directors. C has no power to change the final decision. Hence, A and B have joint control over the entity.

Scenario 2: A does not have control as A needs the support of either B or C to achieve the minimum voting threshold. At the same time, there is more than one possible combination to achieve the 75% threshold. Without any further specification in the contract as to which combination prevails, the situation fails to meet the requirement for unanimous consent by specified investors.[15]

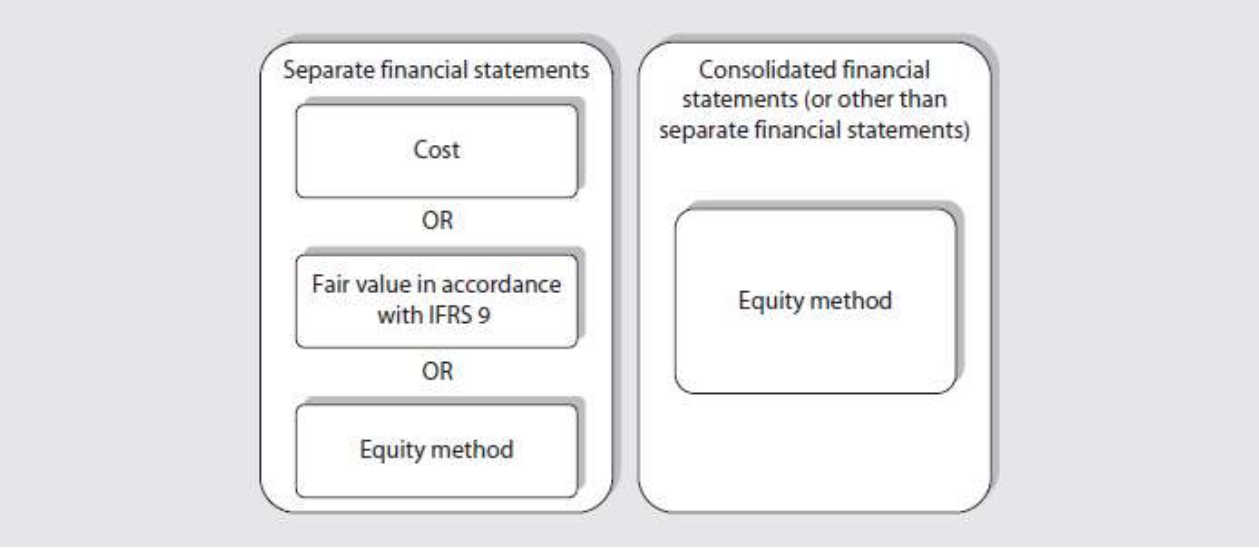
Scenario 3: It is uncertain as to how the other investors will vote. Unanimous consent from A and B is not necessarily the outcome in all situations. In such a situation, A and B have joint control only if the contractual arrangement requires agreement from A and B.[16]

When facts and circumstances change, the investor must re-evaluate the existence of joint control and the nature of the joint arrangement. Changes in ownership interests is an example of such a change and are further discussed in Appendix 7C.

Accounting for Joint Ventures

Joint venturers will account for joint ventures using the equity method in the consolidated financial statements. However, in the separate (legal entity) financial statements, the joint venturer will account for the investment in the joint venture at either cost or as a financial instrument under IFRS 9 or using the equity method. Hence, the accounting for joint ventures (see Figure 6.11) is the same as accounting for associates that we saw earlier in the chapter. The equity method explained earlier in this chapter applies to the accounting of joint ventures in the consolidated financial statements or the investor's financial statements other than separate financial statements. An example of accounting for joint ventures is shown in Illustration 6.4.

FIGURE 6.11 Accounting for joint ventures



Under IAS 27, a joint venturer has a choice of three measurement bases to account for the investment in joint ventures in its separate financial statements. However, the same accounting basis must apply to all joint venture investments. The measurement bases are the same as those, which apply to accounting for investments in subsidiaries and accounting for associates in the separate financial statements of the parent or investor. If a joint venturer chooses to apply the equity method in accounting for investments in joint ventures in the separate financial statements, the investment amount will be the same in both the separate and consolidated financial statements.

ILLUSTRATION 6.4 Accounting for joint arrangements

P Co and T Co acquired a 50% interest each in Z Co on 1 January 20x2. Z Co was incorporated in 20x0. P Co and T Co agreed on the contractual sharing of power to govern the activities of Z Co. P Co does not have rights to the assets of Z Co nor obligations to the liabilities of Z Co. By virtue of its rights to the net assets of Z Co, P Co is a joint venturer of Z Co and applies the equity method to account for its interests in Z Co. The financial statements of P Co and Z Co for the year ending 31 December 20x3 are shown below:

Income Statement and Statement of Changes in Equity (extract)
For Year Ended 31 December 20x3

	P Co	Z Co
Revenues	$12,000,000	$6,000,000
Expenses	(7,000,000)	(5,000,000)
Profit before tax	$ 5,000,000	$1,000,000
Tax	(1,000,000)	(200,000)
Profit after tax	$ 4,000,000	$ 800,000
Dividends declared	(300,000)	(50,000)
Profit retained	$ 3,700,000	$ 750,000
Retained earnings, 1 Jan 20X3	1,200,000	500,000
Retained earnings, 31 Dec 20X3	$ 4,900,000	$1,250,000

Statement of Financial Position
As at 31 December 20x3

	P Co	Z Co
Fixed assets, net book value	$3,500,000	$1,900,000
Other long-term assets	1,800,000	
Investment in Z Co, at cost	600,000	
Amount due from Z Co	60,000	
Inventory	800,000	300,000
Accounts receivable	520,000	400,000
Cash	100,000	20,000
	$7,380,000	$2,620,000

Accounts payable	$ 390,000	$ 710,000
Other liabilities	90,000	
Amount due to P Co		60,000
Share capital	2,000,000	600,000
Retained earnings	4,900,000	1,250,000
	$7,380,000	$2,620,000

	Z Co
Date of investment in joint venture	1 Jan 20x2
Percentage acquired by P Co	50%
Shareholders' equity at date of investment:	
Share capital	$600,000
Retained earnings	200,000
	$800,000

The fair and book values of net assets at date of investment are as follows:

	Z Co	
	Book value	Fair value
Inventory	$200,000	$300,000
Other net assets	600,000	600,000
Total net assets	$800,000	$900,000

The undervalued inventory was resold to third parties during 20x2. Assume a tax rate of 20%. Prepare the equity accounting entries for 20x3 and determine the balance of the investment in the joint venture.

EA1: Recognize share of post-acquisition retained earnings (RE) of Z

Dr Investment in Z	150,000	
Cr Opening RE		150,000

RE of Z as at 1 Jan 20x3	$500,000
RE of Z as at date of investment	200,000
Change in RE	$300,000
Share of Z's change in RE	$150,000

EA2: Recognize past cost of sales of undervalued inventory, after-tax

Dr Opening RE	40,000 (50% × 80% × 100,000)	
Cr Investment in Z		40,000

EA3: Reclassify dividend income as a reduction of investment

Dr Dividend income	25,000	
Cr Investment in Z		25,000

EA4: Recognize share of current profit after tax of Z

Dr Investment in Z	400,000 (50% × 80% × 1,000,000)	
Cr Share of profit of Z		400,000

Analytical check of Investment in Z:

Book value of shareholders' equity of Z			$1,850,000
P's share of Z's identifiable net assets			$ 925,000
Implicit goodwill in investment in Z:			
Investment in Z		$600,000	
BV of net assets of Z at investment date	$800,000		
Excess of FV over BV of inventory (after tax)	80,000		
FV of net assets of Z at investment date	$880,000		
Less Share of FV of net assets of Z at investment		440,000	
Goodwill in Z implicit in the investment in Z			160,000
			$1,085,000
Investment in Z, at cost			$ 600,000
EA1: Recognize share of post-acquisition retained earnings of Z			150,000
EA2: Recognize past cost of sales of undervalued inventory, after tax			(40,000)
EA3: Reclassify dividend income as a reduction of investment			(25,000)
EA4: Recognize share of current profit after tax of Z			400,000
Investment in Z as at 31 Dec 20x3			$1,085,000

Accounting for Joint Operations

Joint operators, on the other hand, will account for their interests in the joint operations in the same manner in both the separate and consolidated financial statements. In both the separate and consolidated financial statements, each operator will "recognize in relation to its interest in a joint operation:

1. Its assets, including its share of any assets held jointly;
2. Its liabilities, including its share of any liabilities incurred jointly;
3. Its revenue from the sale of its output arising from the joint operation;
4. Its share of the revenue from the sale of the output by the joint operation; and
5. Its expenses, including its share of any expenses, incurred jointly."[17]

An example of the accounting for joint operations is shown in Illustration 6.5.

ILLUSTRATION 6.5 Accounting for joint operations

P Co and T Co formed a special purpose entity Z with the sole purpose of acquiring its output of extracted minerals. P Co and T Co agreed on the contractual sharing of power that required unanimous consent on all strategic activities of Z. P Co and T Co fund the activities of Z Co. By virtue of its rights to the assets of Z and its obligations for the liabilities of Z, P Co is a joint operator of Z and recognizes its share of the assets and liabilities of Z on its statement of financial position. P Co and T Co share in Z's assets, liabilities, revenues, and expenses equally.

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The financial statements of P Co for 20x3 result from the summation of P Co's and 50% of Z Co's revenues, expenses, assets and liabilities.

Income Statement and Statement of Changes in Equity (extract)
For Year Ended 31 December 20x3

	P Co (without Z)	Z	P Co (with Z)
Revenues	$6,000,000	$3,000,000	$7,500,000
Expenses	(4,000,000)	(2,000,000)	(5,000,000)
Profit before tax	$2,000,000	$1,000,000	$2,500,000
Tax	(400,000)	(200,000)	(500,000)
Profit after tax	$1,600,000	$ 800,000	$2,000,000
Dividends	(300,000)		(300,000)
Profit retained	$1,300,000	$ 800,000	$1,700,000
Retained earnings, 1 Jan 20X3	1,200,000	500,000	1,450,000
Retained earnings, 31 Dec 20X3	$2,500,000	$1,300,000	$3,150,000

Statement of Financial Position
As at 31 December 20x3

	P Co (without Z)	Z	P Co (with Z)
Fixed assets, net book value	$3,500,000	$1,900,000	$4,450,000
Other long-term assets	500,000		500,000
Initial investment in Z	300,000		–
Inventory	800,000	240,000	920,000
Accounts receivable	520,000	400,000	720,000
Cash	100,000	20,000	110,000
	$5,720,000	$2,560,000	$6,700,000
Current liabilities	$720,000	$200,000	$ 820,000
Long-term liabilities	500,000	460,000	730,000
Initial capital	2,000,000	600,000	2,000,000
Retained earnings	2,500,000	1,300,000	3,150,000
	$5,720,000	$2,560,000	$6,700,000

The share of revenue, expenses, assets, and liabilities are recognized period-by-period in the books of the joint operator. Hence, there is no need to "re-enact" prior year entries. There is also no difference between legal entity and economic entity reporting for joint operations.

To illustrate the journal entries that are passed in the P Co's books, we show the statement of financial position of Z as at 31 December 20x2 below.

Statement of Financial Position as at 31 December 20x2

Fixed assets, net book value	950,000
Inventory	320,000
Accounts receivable	150,000
Cash	50,000
	1,470,000

Current liabilities	130,000
Long-term liabilities	240,000
Initial capital	600,000
Retained earnings	500,000
	1,470,000

Journal entry by P Co during 20x3

Dr Expenses	1,000,000		
Dr Tax	100,000		
Dr Fixed assets	475,000	50% × (1,900,000−950,000)	
Dr Accounts receivable	125,000	50% × (400,000−150,000)	
Cr Cash		15,000	
Cr Inventory		40,000	
Cr Current liabilities		35,000	50% × (200,000–130,000)
Cr Long-term liabilities		110,000	50% × (460,000–240,000)
Cr Revenue		1,500,000	

Recognition of assets, liabilities, revenue, and expenses during 20x3

In this example, because there is an equal sharing of 50% across all line items, the share of equity attributable to P Co is equal to the share of net assets attributable to P Co.

Share of initial share capital	300,000
Share of opening retained earnings	250,000
Share of revenue	1,500,000
Share of expenses	(1,000,000)
Share of tax	(100,000)
Share of equity attributable to P Co	950,000

Share of fixed assets	950,000
Share of inventory	120,000
Share of accounts receivable	200,000
Share of cash	10,000
Share of current liabilities	(100,000)

Share of long-term liabilities	(230,000)
Share of net assets attributable to P Co	950,000

Where the proportion shared across revenues, expenses, assets, and liabilities is not constant, the share of equity attributable to a joint operator will not be equal to the share of net assets attributable to the joint operator. The contractual arrangement will need to spell out how shortfalls are dealt with. Typically, the other joint operator with the excess will reimburse the joint operator with the shortfall.

ILLUSTRATION 6.6 Accounting for joint operations

On 1 January 20x3, P Co and T Co entered into a joint arrangement to extract mineral oil in deep ocean. P Co contributes equipment and T Co the mining rights and mining expertise. Assume that the useful life of the equipment is 50 years. The revenues and expenses are shared equally. Expenses include the depreciation of equipment, amortization of mining rights and employee expenses. At 31 December 20x3, the year end, the following amounts were recorded:

Equipment, carrying amount in P Co's books	US$200,000,000
Mining rights, carrying amount in T Co's books	US$100,000,000
Revenue from sale of mineral oil	US$35,000,000
Expenses incurred in 20x3	US$15,000,000
Output of oil distributed to each operator	1,000,000 barrels of oil (Market price is US$40 per barrel)

In this example, there is no separate vehicle for the joint operation. T Co is the project manager and has custody of the bank account.

P Co and T Co will recognize the equipment and mining rights separately in their own books. Share of revenue, expenses and value of the physical output will be recognized by each operator.

The journal entries passed during 20x3 by P Co are as follows:

Dr Project Account	US$4,000,000	
Cr Accumulated depreciation		US$4,000,000
Depreciation expense on equipment loaned to project		
Dr Depreciation expense	US$2,000,000	
Dr Other expense	US$5,500,000*	
Cr Project Account		US$7,500,000
Share of expenses on project (including share of depreciation expense)		

* (US$15,000,000 - US$4,000,000)/2

Dr Project Account	US$17,500,000	
Cr Revenue		US$17,500,000
Share of revenue on project (US$35,000,000/2)		
Dr Inventory	US$40,000,000	
Cr Revenue		US$40,000,000
Share of physical output from project (1,000,000 × US$40)		
Dr Cash	US$14,000,000	
Cr Project Account		US$14,000,000
Cash received from T Co in settlement of project account		

The project account is settled by cash. P Co as a joint operator receives cash from the project and physical output.

APPENDIX 6A

Sale or Contribution of Assets or Business Between Investor and its Associate

The discussions in Chapter 6 have focused on the sale of assets to and from the investor. It should be noted that, in the case of downstream transactions, paragraph 28 of IAS 28 specifically includes contribution of assets from the investor to its associate other than sales to an associate. Such contributions of assets by the investor could take place at the inception where the investor transfers non-monetary assets to a newly setup associate as part of its initial investment in exchange for the equity instruments issued by the associate. It can also take place post-inception where such assets are contributed in exchange for additional equity investments.

Paragraph 30 of IAS 28 requires the contribution of a non-monetary asset to an associate in exchange for an equity interest in the associate to be accounted for in accordance with paragraph 28 of IAS 28 except when the contribution lacks commercial substance[18], i.e. it requires the gains or losses resulting from the contribution of assets to be recognized in the investor's financial statements only to the extent of the unrelated investor's interest in the associate. As discussed earlier, what this effectively means is that the investor's share of the gain or loss is eliminated as it is deemed to be unrealized.

However, paragraph 30 deals only in situations where non-monetary assets are contributed. In some cases, the investor may contribute its interests in a subsidiary to its associate. From the investor's perspective, the investor would have lost control of the subsidiary at the time when it contributes the interests in the subsidiary to the associate.

Consequently, the investor will have to account for that loss of control in accordance with the requirements in IFRS 10.

Prior to the issuance of the Amendment to IFRS 10 and IAS 28 *Sale or Contribution of Assets between an Investor and its Associate or Joint Venture*[19], a conflict existed between the requirements of IFRS 10 relating to the accounting for loss of control of a subsidiary and the requirements of IAS 28 pertaining to the accounting for contribution of assets by an investor. In particular, the pre-amended paragraph 25 of the IFRS 10 requires an entity that loses control of a subsidiary but retains an interest (which is now accounted for as an associate if it retains significant influence) to account for that retained interest at fair value on the date where control is lost and that is included in calculating the gain or loss associated with the loss of control of the subsidiary. This fair value is regarded as the cost on initial recognition of an investment in an associate for which equity accounting will apply for subsequent measurement. Paragraph 28 of IAS 28, on the other hand, requires an investor to restrict the recognition of the gain or loss arising from the contribution or sale of non-monetary assets to the associate only to the extent of unrelated interests. This requirement runs counter to the requirement in IFRS 10 where no restriction is required in respect of the gain or loss to the extent of unrelated interest.

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Hence, in order to rectify this, the Board issued the said Amendments above in September 2014. The Amendments, specifically B99A of IAS 28 requires a parent to calculate the gain or loss arising from the loss of control of a subsidiary that does not contain a business arising from a transaction[20] with an associate or joint venture in accordance with B98 to B99[21] of IFRS 10. The parent will recognize the gain or loss in its profit and loss only to the extent of the unrelated investors' interest in the associate or joint venture. The remaining amount is eliminated against the carrying value of the associate.

In addition, if the parent as part of the transaction, retains an investment in the former subsidiary such that it is now an associate or a joint venture that is equity accounted for by the parent, the parent which is required to recognize the retained interest at fair value, will only recognize such remeasurement gains or losses to the extent of the unrelated investors' interests in this associate or joint venture. The remaining part is eliminated against the carrying value of the investment in associate retained in the parent. However, if the retained interest is accounted for in accordance with IAS 39/IFRS 9, the part of the gain or loss arising from the remeasurement of the retained interest at fair value is recognized in full in the parent's profit and loss[22].

Conversely, if the subsidiary in which the parent lost control over constitutes a business under the definition of IFRS 3, the full gain or loss calculated in accordance with B98 to B99 of IFRS 10 is recognized by the parent. Similarly, any interest retained in this former subsidiary is recognized at fair value in full in the parent's financial statements.

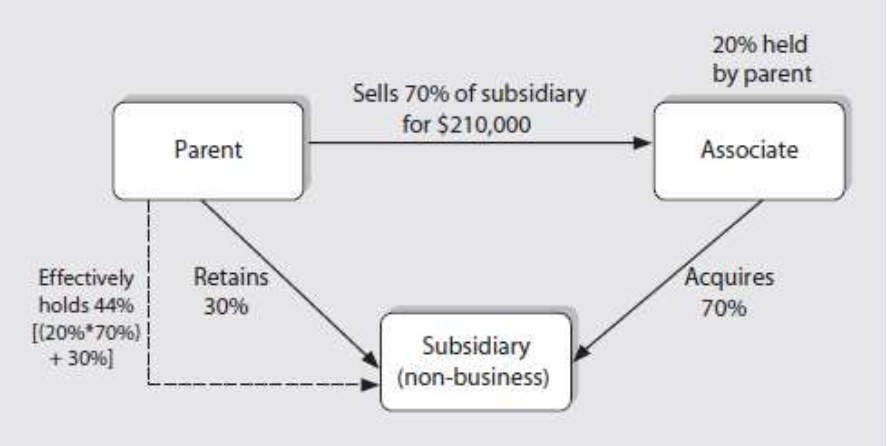
In developing the Amendments, the Board's primary concern was centered around the possibility that the existing requirements in IFRS 10 and IAS 28 could result in different outcomes depending on how the transaction is structured, i.e. it could result in the accounting for a transaction being driven by its form rather than substance. For instance, different accounting might be applied depending on whether the assets were transferred in a transaction that was structured as a sale of assets or as a sale of the entity that holds the assets (BC 190D to IAS 28). However, the Board concluded that on balance, for consistency purposes, the accounting for loss of control of a business should be consistent with the conclusions in IFRS 10 and hence, a full gain or loss should be recognized on loss of control regardless of whether the business is housed in a subsidiary or not. As assets that do not constitute business were not part of the Board's Business Combination project on which IFRS 3 was subsequently issued, the Board concluded that the current requirements in IAS 28 on the partial recognition of gains or losses in transactions between the investor and associate should only apply to sale or contribution of assets that do not constitute a business under IFRS 3 (BC190E, BC190F to IAS 28).

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The following illustration which is adapted from the application example 17 in the Amendments demonstrates the accounting above.

ILLUSTRATION 6A.1 Accounting for sale of subsidiary that does not contain a business to an associate

A parent has a 100% interest in a subsidiary that does not contain a business. The parent sells 70% of its interest in the subsidiary to an associate in which it has 20% interest. As a result of this transaction, the parent loses control of the subsidiary, but retained 30% equity interest after the transaction. This 30% interest retained in the former subsidiary is equity accounted for as an associate in parent's financial statements. The fair value of the consideration received from the associate is $210,000 is also equivalent to the fair value of the 70% interest sold. The cost of investment of subsidiary in the separate financial statements of parent is $50,000. The following also information is relevant at the date of the transaction.



	Carrying Values of Net Assets $'000	Fair Value $'000
Subsidiary	100	300

Analysis

Calculation of the gain on disposal in the consolidated financial statements of parent prior to elimination required by B99A of IFRS 10

Gain on disposal	Fair value of consideration	+	Fair value of retained interest at date when control is lost	–	Carrying value of net assets[23] at date when control is lost
	= 210,000	+	90,000*	–	100,000
	= 200,000				

* $90,000 = 30% interest held * Fair value of investment of $300,000.

The gain on disposal above comprises of two components as follows:

Gain on disposal	=	Gain on sale of 70% of subsidiary to associate	+	Gain on remeasurement of retained interest of 30% to fair value
	=	(Fair value of consideration – carrying value of 70% interest)	+	(Fair value of 30% retained interest – Carrying value of 30% retained interest)
	=	(210,000 – 70%*100,000)	+	(30%*300,000 – 30%*100,000)
	=	140,000	+	60,000
	=	200,000		

Calculation of the adjusted gain on disposal in the consolidated financial statements of parent and amount to be eliminated against cost of investment in associate as required by B99A of IFRS 10

Gain on disposal recognized by parent – post elimination	=	Gain on sale of 70% of subsidiary to associate*Equity shareholding held by unrelated interest	+	Gain on remeasurement of retained interest of 30% to fair value*Equity shareholding held by unrelated interest
	=	140,000*80% interest	+	60,000*56% interest
	=	112,000	+	34,000
	=	146,000		

Hence, the amount to be eliminated against cost of investment in associate

Amount to be eliminated against cost of investment in associate	=	Gain on sale of 70% of subsidiary to associate*Equity shareholding held by parent	+	Gain on remeasurement of retained interest of 30% to fair value*Equity shareholding held by parent
	=	140,000*20% interest	+	60,000*44% interest
	=	28,000	+	26,000
	=	54,000		

Calculation of gain on disposal in parent's separate financial statements

Gain on disposal	=	Sales proceeds	–	Share of cost of investment disposed
	=	Sales proceeds	–	70%*Cost of investment
	=	210,000	–	70%*50,000
	=	210,000	–	35,000
	=	175,000		

Preparation of accounting entries

In the consolidated financial statements

Dr Gain on disposal	35,000[#]	
Cr Opening retained earnings		35,000

Being adjusted gain on disposal at group level prior to elimination as required by B99A

[#] 35,000 = 70%* Post acquisition profits ($100,000 – $50,000)

Dr Gain on disposal	28,000	
Cr Cost of investment in associate		28,000

Being recognition of gain on disposal to the extent of unrelated interest, i.e. 80%

Dr Cost of investment in former subsidiary, now associate	60,000	
Cr Gain on disposal		60,000

Being remeasurement of retained interest to fair value prior to elimination as required by B99A

Dr Gain on disposal	26,000	
Cr Cost of investment in former subsidiary, now associate		26,000

Being restraining the gain on disposal to the extent of unrelated party interests in the former subsidiary now associate

In the parent's separate financial statements

Dr Cash	210,000	
Cr Cost of investment		35,000
Cr Gain on disposal		175,000

Being gain on disposal on sale of former subsidiary in the separate financial statements In the associate's financial statements

Dr Net Assets	210,000[+]	
Cr Cost of investment		210,000

Being accounting for acquisition of assets in the associate's financial statements

[+]In the associate's financial statements, this sale of the subsidiary that does not constitute a business is accounted for as an acquisition of assets. The purchase consideration is allocated to the individual assets and liabilities in the subsidiary based on the relative fair values at the date of the transaction.

Conversely, if the scenario were to be tweaked such that the subsidiary sold by the parent constitutes a business, the gain on disposal of $200,000 calculated above will be recognized in full in the consolidated financial statements of the parent. The associate will account for the transaction as a business combination under IFRS 3.

Subsequent to the issuance of the Amendments to IFRS 10 and IAS 28 *Sale or Contribution of Assets between an Investor and its Associate or Joint Venture,* an unintended consequence was uncovered. As discussed above, when the parent retains an interest in the transaction described in B99A of IFRS 10, this interest is remeasured to fair value at the date when control is lost. The gain or loss on the remeasurement recognized by the parent in its consolidated financial statement is restricted to the extent of the unrelated investors' interests in the associate and the remaining part of the gain or loss is eliminated against the carrying amount of the investment in that associate. Consequently, the net carrying value (i.e. fair value adjusted for the gain or loss of the parent's share in the now associate) forms the cost of investment in the associate on acquisition.

However, paragraph 32 of IAS 28 provides guidance on the accounting for the difference between the cost of the investment recognized on acquisition of the investment and the investor's share of net assets of the equity accounted for associate. Specifically, goodwill (i.e. the excess of cost of investment over the investor's share of fair value of net assets) is included in the carrying amount of the investment and any excess of the investor's share of fair value of net identifiable assets over the cost of investment is included as income in the determination of the investor's share of associate's profit or loss in the period in which the investment is acquired.

As the gain or loss of remeasurement to the extent of parent's interest is eliminated against the cost of investment as required in B99A of IFRS 10, there is a concern that the carrying value of the cost of investment may be reduced significantly when the remeasurement of the retained interest in the associate results in a gain and when the investor's share of fair value of the net identifiable assets is calculated in applying the requirements of paragraph 32 of IAS 28, this may result in an excess of the investor's share in the net assets over the cost of investment for which the excess is required to be recognized as a gain in the investor's profit and loss. Effectively in this case, the gain arising from the remeasurement which was previously eliminated against the cost of investment in accordance with paragraph B99A of IFRS 10 is re-recognized as income for the period, thereby completely reversing the elimination of the gain required under paragraph B99A of IFRS 10.

The Board deliberated in January 2015 and tentatively agreed to clarify the requirements of IFRS 10 and IAS 28 to:

1. Amend IFRS 10 to explain that in the limited circumstance when the loss of control of a subsidiary gives rise to an investment that is equity accounted for and that subsidiary does not contain a business, the cost on initial recognition of the investment is the fair value of the investment; and any gains or losses eliminated are a subsequent adjustment and
2. Amend IAS 28 to explain that for the limited circumstance described above, the cost of acquisition of the associate for the purpose of acquisition accounting is the fair value of the investment at the date where control is lost and is determined before the elimination of any gains or losses required by paragraph B99A of IFRS 10.

However, in June 2015, the Board decided to consider the above tentative decisions together with other proposals relating to application of IAS 28 as part of a wider research project on the equity method of accounting. At the same time, they also tentatively decided to defer the effective date of the Amendments to IFRS 10 and IAS 28 so that entities would not need to change the way in which they apply IAS 28 twice in a short period. In addition, the Board decided that early application of the Amendment to IFRS 10 and IAS 28 should continue to be permitted because the amendments were intended to address existing diversity in practice and consequently, the IASB was of the view that early application of the Amendments to IFRS 10 and IAS 28 was unlikely to increase diversity. An exposure draft (ED/2015/17) on these proposals was issued in August 2015 to seek constituents' views on the proposals. Majority of the respondents agreed with the proposal and the Board finalized the deferral of the September 2014 Amendment in December 2015. The original effective date of 1 January 2016 was deleted and the IASB indicated that a new effective date will be determined at a future date when the Board finalizes the revisions, if any that result from the research project on the equity method of accounting.

PROBLEMS

P6.1 Equity method versus consolidation

Refer to the information in P5.2.

Required:

1. If Jewel could legitimately structure the relationship with Opal such that it could equity account rather than consolidate Opal's financial statements, what would be the reported balance in the investment in Opal account as at 31 December 20x2?

2. Advise Jewel on whether, and how, investors may differ in their perceptions of the risk and value of the economic entity as a result of the different information presented under equity as compared with consolidation for group reporting.

P6.2 Equity method and analytical check on Investment in Associate

Refer to the information in P5.3.

Required:

1. If other evidence suggests that Prism had only significant influence and not control over Sapphire, prepare the necessary equity accounting entries for the year ended 31 December 20x9.
2. Perform an analytical check on the balance of the investment in Sapphire account under the equity method.

P6.3 Analysis of cost method versus equity method

On 1 January 20x2, I-Co entered into a joint-venture agreement to incorporate a new company called A-Co. I-Co's initial investment in A-Co was $10,000,000, which was 30% of the initial start-up capital. The following information relates to A-Co for the year ended 31 December 20x2:

Profit before tax	$6,000,000
Tax	(1,500,000)
Profit after tax	$4,500,000
Dividends	(150,000)
Retained profit	$4,350,000

Required:

1. Assume that if I-Co had a choice between the cost and equity method, what amounts would be reported for the following items under either method?
   (a) The after-tax income from A-Co for the year ended 31 December 20x2; and
   (b) The carrying amount of A-Co as at 31 December 20x2.
2. Consider the arguments put forward by V. Mazay, T. Wilkins and I. Zimmer, "Determinants of the Choice of Accounting for Investments in Associated Companies", *Contemporary Accounting Research*, Vol. 10 No. 1 (Fall 1993). Their paper focuses on the motivations for the choice between cost and the equity accounting methods. Using insights from the paper and your own critical analysis, explain which of the two methods would be preferred by the following groups and under what conditions. page 394
   (a) I-Co's management;
   (b) I-Co's equity holders; and
   (c) I-Co's lenders.

A note on non-elimination of profits arising from transactions with associates by Mazay et al.: When an investor sells to an associate or subsidiary, a profit may be earned and recorded in the books of the investor. The profit arises when the transfer price is at a mark-up to the original cost incurred by the investor. The associate or subsidiary that buys the goods will then sell to an external party and record a cost of sale based on the transfer price. If the two transactions (the original sale and the resale) are made within the same period, there is no issue with any "unrealized profit" as the sales recorded by the investor will be offset by the cost of sales recorded by the associate. However, when the two sales are not in the same period, an unrealized profit arises from an intragroup transaction. From the group's perspective, this profit does not exist as it arises from a sale to a related party within the group. It

is also a result of intragroup transfer pricing, which has not been ratified by an external market transaction. (The above arguments also hold when the associate or subsidiary sells to the investor.) Unrealized profits must be adjusted out in *equity accounting* and *consolidation*. The *cost method*, however, emphasizes the legal entity perspective: the investor and associate are deemed as two unrelated parties and the "unrealized" profit arising from the sale is deemed as earned.

P6.4 Equity method

The financial statements of A Co are shown below. P Co acquired a 30% interest in A Co on 1 January 20x3.

Income Statement
For Year Ended 31 December 20x5

Profit before tax	$200,000
Tax at 20%	(40,000)
Profit after tax	$160,000
Dividends declared	(20,000)
Profit retained	$140,000
Retained earnings, 1 January 20x5	100,000
Retained earnings, 31 December 20x5	$240,000

Statement of Financial Position
As at 31 December 20x5

Fixed assets, net book value	$450,000
Inventory	120,000
Accounts receivable	60,000
Cash	25,000
	$655,000
Accounts payable	$315,000
Share capital	100,000
Retained earnings	240,000
	$655,000
Date of acquisition of A Co	1 January 20x3
Percentage acquired by P Co	30%
Cost of investment in A Co	$200,000
Shareholders' equity of A Co at date of acquisition:	
Share capital	100,000
Retained earnings	30,000
	$130,000

Fair and book values of identifiable net assets at date of acquisition:	
Unrecognized intangible asset (fair value)	$ 40,000
Other identifiable net assets (book and fair values) .	130,000
Fair value of identifiable net assets at date of acquisition	170,000
Book value of identifiable net assets at date of acquisition	130,000

Additional information:

(a) Unrecognized intangible asset of A Co was partially impaired in 20x4 to the extent of 50% of its original fair value.

(b) A Co purchased inventory from P Co on 1 December 20x4. Details are as follows:

Transfer price	$20,000
Original cost	(15,000)
Profit on sale	$ 5,000

The inventory was sold only in 20x5.

(c) Assume a tax rate of 20%. Recognize tax on fair value adjustments.

Required:

1. Prepare the equity accounting entries for 20x5.
2. Perform an analytical check on the balance of Investment in A Co as at 31 December 20x5.

P6.5 Comprehensive problem set

On 1 January 20x1, P Co acquired 70% of S Co by issuing 1,000,000 new shares to the owners of S Co. The fair value of consideration paid by P Co to acquire S Co was $2,100,000. The fair value of non-controlling interests at acquisition date was $900,000. On 1 January 20x2, P Co acquired a 40% ownership interest in A Co by making a cash payout of $200,000 to the owners of A Co.

The following information relates to statements of financial position at date of acquisition:

	S Co Book value	S Co Fair value	A Co Book value	A Co Fair value
In-process R&D	$ 0	$1,000,000		
Fixed assets	50,000	50,000	$100,000	$140,000
Inventory	500,000	550,000	120,000	120,000
Accounts receivable	100,000	100,000	140,000	140,000
Cash	20,000	20,000	10,000	10,000
Total assets	$670,000	$1,720,000	$370,000	$410,000
Contingent liability	$ 0	$ 50,000		
Accounts payable	250,000	250,000	$ 50,000	$ 50,000
Share capital	300,000		200,000	
Retained earnings	120,000		120,000	
Equity and Liabilities	$670,000	$1,720,000	$370,000	$410,000

The following financial statements relate to the financial year ended 31 December 20x3.

Income statement
For Year Ended 31 December 20x3

	P Co	S Co	A Co
Sales	$5,000,000	$3,500,000	$2,000,000
Cost of sales	(4,000,000)	(3,000,000)	(1,300,000)
Gross profit	$1,000,000	$ 500,000	$ 700,000
Other income	140,000	30,000	20,000
Operating expenses	(150,000)	(70,000)	(100,000)
Net profit before tax	$ 990,000	$ 460,000	$ 620,000
Tax	(198,000)	(92,000)	(136,400)
Net profit after tax	$ 792,000	$ 368,000	$ 483,600
Dividends declared	(50,000)	(100,000)	(60,000)
Net profit attributable to shareholders	$ 742,000	$ 268,000	$ 423,600
Retained earnings, 1 January 20x3	2,500,000	250,000	160,000
Retained earnings, 31 December 20x3	$ 3,242,000	$ 518,000	$ 583,600

Statement of Financial Position
As at 31 December 20x3

	P Co	S Co
Investment in S Co	$2,100,000	
Investment in A Co	200,000	
Fixed assets	4,000,000	$ 40,000
Inventory	2,000,000	450,000
Accounts receivable	1,000,000	350,000

Cash	120,000	40,000
	$9,420,000	$880,000
Accounts payable	$3,178,000	$ 62,000
Share capital	3,000,000	300,000
Retained earnings	3,242,000	518,000
	$9,420,000	$880,000

Additional information:

(a) Impairment of in-process research and development of $100,000 was expensed off in the consolidated financial statements of P Co in 20x2.

(b) Undervalued inventory as at 1 January 20x1 was sold in 20x1.

(c) The contingent liability of $50,000 in respect of legal claims was paid off by S in 20x3, and recognized as an expense by S Co in 20x3.

(d) The following sales of inventory were made during 20x2 and 20x3.

	20x2	20x3
Sales from S to P	$100,000	
Original cost	$ 80,000	
Percentage unsold to third parties at year-end	20%	0%
Sales from S to P		$200,000
Original cost		$160,000
Percentage unsold to third parties at year-end		40%
Sales from P to A	$ 70,000	
Original cost	$ 50,000	
Percentage unsold to third parties at year-end	30%	0%

(e) Undervalued fixed assets of A Co had a useful life of five years from date of acquisition.

(f) There is no change in the share capital of S Co and A Co from acquisition date.

(g) Assume a tax rate of 20%. Recognize tax on fair value adjustments.

Required:

1. Prepare the consolidation and equity accounting entries for the year ended 31 December 20x3.
2. Prepare the consolidation worksheet for the year ended 31 December 20x3 to show the consolidated financial statements prepared in accordance with IAS 27 and IAS 28.
3. Perform an analytical check on the balance of non-controlling interests and investment of associate as at 31 December 20x3.

P6.6 Comprehensive problem set

P Co acquired an interest in X Co and Z Co. Details of the acquisitions are as follows:

	X Co	Z Co
Date of acquisition	1 Jan 20x3	1 Jan 20x4
Percentage acquired by P Co	90%	
Percentage acquired by P Co		30%
Shareholders' equity at date of acquisition:		
Share capital	$ 500,000	$ 300,000
Retained earnings	1,000,000	800,000
	$1,500,000	$1,100,000

The fair value and book value of assets at acquisition date are shown below:

	X Co Book value	X Co Fair value	Z Co Book value	Z Co Fair value
Inventory	$ 100,000	$ 150,000		
Intangibles				$ 200,000
Other identifiable net assets	1,400,000	1,400,000	$1,100,000	1,100,000
Total identifiable net assets	$1,500,000	$1,550,000	$1,100,000	$1,300,000
Fair value of non-controlling interests		$ 200,000		

The financial statements for the year ended 31 December 20x5 are shown below:

Income Statement
For Year Ended 31 December 20x5

	P Co	X Co	Z Co
Profit before tax	$8,000,000	$5,000,000	$1,200,000
Tax	(1,600,000)	(1,000,000)	(200,000)
Profit after tax	$6,400,000	$4,000,000	$1,000,000
Dividends declared	(600,000)	(240,000)	(100,000)
Profit retained	$5,800,000	$3,760,000	$ 900,000
Retained earnings, 1 January 20x5	1,800,000	1,600,000	1,000,000
Retained earnings, 31 December 20x5	$7,600,000	$5,360,000	$1,900,000

Statement of Financial Position
As at 31 December 20x5

	P Co	X Co	Z Co
Fixed assets, net book value	$5,000,000	$4,850,000	$2,000,000
Investment in X, at cost	1,800,000		
Investment in Z, at cost	500,000		
Inventory	750,000	600,000	200,000
Accounts receivable	700,000	600,000	300,000

Cash	200,000	150,000	50,000
	$8,950,000	$6,200,000	$2,550,000
Accounts payable	$ 350,000	$ 340,000	$ 350,000
Share capital	1,000,000	500,000	300,000
Retained earnings	7,600,000	5,360,000	1,900,000
	$8,950,000	$6,200,000	$2,550,000

Additional information:

(a) Undervalued inventory was sold in 20x4.

(b) Unrecognized intangible asset of Z Co was impaired in 20x5 to the extent of 20% of its original fair value. This was unrecognized in the separate financial statements.

(c) X Co transferred its fixed asset to P Co on 1 January 20x4. Details are as follows:

Transfer price invoiced by X Co to P Co		$120,000
Original cost of the fixed asset	$100,000	
Accumulated depreciation	60,000	
Net book value at 1 January 20x4		40,000
Profit on sale recorded by X Co in 20x4		$ 80,000
Useful life in years from date of purchase		5

Assume no change in estimates of useful life, i.e. remaining life at 1 January 20x4 is two years. Also assume nil residual value.

(d) X Co purchased inventory from P Co on 15 December 20x5. Details are as follows:

Transfer price	$30,000
Original cost	10,000
Gain on sale recorded by P Co	$20,000

The inventory was sold only in 20x6

(e) Sales between P Co and Z Co are as follows:

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	Sales price	Original cost	Unsold at end of transfer year
P Co sold to Z Co in 20x4	$100,000	$80,000	10%
Z Co sold to P Co in 20x5	60,000	45,000	20%

The inventory sold by P Co to Z Co was fully resold to third parties in 20x5.

(f) Assume a tax rate of 20%. Recognize tax on fair value adjustments.

Required:

1. Prepare the consolidation and equity accounting entries for the year ended 31 December 20x5.
2. Perform an analytical check on the non-controlling interests' balance as at 31 December 20x5.
3. Perform an analytical check on the investment in associate balance as at 31 December 20x5.

P6.7 Comprehensive problem set

On 1 January 20x3, P Co acquired a 90% interest in Y Co. On that date, the fair value of non-controlling interests in Y Co was $180,000. A year later, on 1 January 20x4, P Co acquired a 30% interest in Z Co. Book values of equity and fair values of identifiable net assets of the acquired companies are shown below:

	Y Co	Z Co
Date of acquisition	1 January 20x3	1 January 20x4
Shareholders' equity at date of acquisition:		
Share capital	$ 500,000	$ 300,000
Retained earnings	1,000,000	700,000
	$1,500,000	$1,000,000

The fair and book values of identifiable net assets of each company at date of acquisition are shown below:

	Y Co		Z Co	
	Book value	Fair value	Book value	Fair value
Intangibles	$ 0	$ 200,000	$ 0	$ 300,000
Other identifiable net assets	1,500,000	1,500,000	1,000,000	1,000,000
Total identifiable net assets	$1,500,000	$1,700,000	$1,000,000	$1,300,000

The financial statements for the year ended 31 December 20x5 are shown below:

Income Statement
For Year Ended 31 December 20x5

	P Co	Y Co	Z Co
Profit before tax	$5,000,000	$1,300,000	$1,400,000
Tax	(1,000,000)	(260,000)	(300,000)
Profit after tax	$4,000,000	$1,040,000	$1,100,000
Dividends declared	(600,000)	(200,000)	(100,000)
Profit retained	$3,400,000	$ 840,000	$1,000,000
Retained earnings, 1 January 20x5	4,000,000	1,500,000	900,000
Retained earnings, 31 December 20x5	$7,400,000	$2,340,000	$1,900,000

Statement of Financial Position
As at 31 December 20x5

	P Co	Y Co	Z Co
Fixed assets, net book value	$5,000,000	$2,300,000	$1,900,000
Investment in Y Co, at cost	1,800,000		
Investment in Z Co, at cost	500,000		

Inventory	750,000	500,000	200,000
Accounts receivable	1,500,000	150,000	300,000
Cash	50,000	100,000	50,000
	$9,600,000	$3,050,000	$2,450,000
Accounts payable	$1,200,000	$ 210,000	$ 250,000
Share capital	1,000,000	500,000	300,000
Retained earnings	7,400,000	2,340,000	1,900,000
	$9,600,000	$3,050,000	$2,450,000

Additional information:

(a) Unrecognized intangible asset of Z Co was impaired in 20x5 to the extent of 30% of its original fair value. Unrecognized intangible asset of Y Co was unimpaired.

(b) On 1 January 20x5, Y Co sold equipment to P Co at an invoiced price of $150,000. At the date of the sale, the net book value of the equipment was $48,000. Its original cost was $120,000. The original useful life of the equipment was five years; it had no estimated residual value. On 1 January 20x5, the remaining useful life was estimated at three years; estimated residual value remains at nil value.

(c) P Co sold inventory to Y Co in December 20x4 at market price of $60,000. The original cost of the inventory was $70,000. The inventory was resold to third parties in 20x5.

(d) Assume a tax rate of 20%. Recognize tax on fair value adjustments.

Required:

1. Prepare the consolidation and equity accounting entries for the year ended 31 December 20x5 (with narratives and workings).
2. Perform analytical checks on the following balances as at 31 December 20x5:
   (a) Non-controlling interests; and
   (b) Investment in associate.

P6.8 Comprehensive problem set

P Co acquired a 90% ownership interest in Y Co on 1 January 20x3. At the date of acquisition, the share capital of Y Co was $1,000,000, and the retained earnings balance was $500,000. The book values of assets of Y Co were close to their fair values, with except for inventory that was undervalued by $100,000. Y Co sold the undervalued inventory in 20x3. The fair value of non-controlling interests was $200,000 as at the date of acquisition.

On 1 January 20x4, P Co acquired a 30% ownership interest in Z Co. At that date, the share capital of Z Co was $200,000 while its retained earnings balance was $400,000. Z Co had an unrecognized intangible asset with a reliable fair value of $240,000. The intangible asset had a useful life of five years from the date of acquisition.

In 20x4, Y Co sold inventory to P Co at a transfer price of $360,000. The original cost of the inventory was $200,000. Ninety percent of the inventory remained unsold at the end of 20x4; 10% of the original inventory remained unsold as at the end of 20x5.

On 1 January 20x5, Z Co transferred machinery to P Co at an invoiced price of $296,000. The original cost of the machinery was $360,000; the profit on sale recorded by Z Co from the transfer was $8,000. The machinery had an original useful life of five years with no residual value. As at 1 January 20x5, the remaining useful life of the machinery was four years with no change to its estimated residual value.

Assume a tax rate of 20%. Recognize tax effects of fair value adjustments. The financial statements for P Co, Y Co and Z Co for the financial year ended 31 December 20x5 are shown below.

Income Statement
For Year Ended 31 December 20x5

	P Co	Y Co	Z Co
Profit before tax	$3,000,000	$1,200,000	$ 600,000
Tax	(600,000)	(240,000)	(100,000)
Profit after tax	$2,400,000	$ 960,000	$ 500,000
Dividends declared	(300,000)	(200,000)	(60,000)
Profit retained	$2,100,000	$ 760,000	$ 440,000
Retained earnings, 1 January 20x5	1,000,000	900,000	560,000
Retained earnings, 31 December 20x5	$3,100,000	$1,660,000	$1,000,000

Statement of Financial Position
As at 31 December 20x5

	P Co	Y Co	Z Co
Fixed assets, net book value	$3,200,000	$1,500,000	$ 800,000
Investment in Y Co, at cost	1,900,000		
Investment in Z Co, at cost	600,000		
Inventory	800,000	600,000	300,000
Intercompany receivable		200,000	
Accounts receivable	530,000	300,000	250,000
Cash	20,000	80,000	30,000
	$7,050,000	$2,680,000	$1,380,000
Accounts payable	$2,750,000	$ 20,000	$ 180,000
Intercompany payable	200,000		
Share capital	1,000,000	1,000,000	200,000
Retained earnings	3,100,000	1,660,000	1,000,000
	$7,050,000	$2,680,000	$1,380,000

Required:

1. Prepare the consolidation and equity accounting entries for 20x5 (with narratives and workings).
2. Prepare a consolidation worksheet for the year ended 31 December 20x5 for the above companies.
3. Perform an analytical check on the year-end balances of non-controlling interests and investment in associate.

P6.9 Comprehensive problem set

P Co acquired an interest in Y Co and Z Co. Details of the acquisitions are as follows:

	Y Co	Z Co
Date of acquisition	1 January 20x4	1 January 20x5
Percentage acquired by P Co	90%	30%
Shareholders' equity at date of acquisition:		
Share capital	$ 500,000	$200,000
Retained earnings	700,000	400,000
	$1,200,000	$600,000

The differences between fair value and book value at acquisition date were for the following assets:

	Y Co		Z Co	
	Book value	Fair value	Book value	Fair value
Inventory	$220,000	$320,000	–	
Intangibles				$300,000

Inventory of Y Co at acquisition date was sold to third parties within six months of acquisition. Intangible assets of Z Co had a remaining useful life of five years from the date of acquisition. The fair value of non-controlling interests of Y Co at acquisition date was $140,000.

The financial statements of P Co, Y Co, and Z Co for the year ended 31 December 20x6 are shown below.

Income Statement
For Year Ended 31 December 20x6

	P Co	Y Co	Z Co
Profit before tax	$2,800,000	$1,200,000	$ 500,000
Tax	(560,000)	(240,000)	(80,000)
Profit after tax	$2,240,000	$ 960,000	$ 420,000
Dividends declared	(200,000)	(120,000)	(100,000)
Profit retained	$2,040,000	$ 840,000	$ 320,000
Retained earnings, 1 January 20x6	2,800,000	800,000	700,000
Retained earnings, 31 December 20x6	$4,840,000	$1,640,000	$1,020,000

Statement of Financial Position
As at 31 December 20x6

	P Co	Y Co	Z Co
Fixed assets, net book value	$2,340,000	$1,000,000	$ 700,000
Investment in Y Co, at cost	1,400,000		
Investment in Z Co, at cost	500,000		
Inventory	560,000	550,000	230,000
Intercompany receivable		40,000	
Amount owing by P Co			50,000

Accounts receivable	1,200,000	800,000	300,000
Cash	30,000	100,000	50,000
	$6,030,000	$2,490,000	$1,330,000
Accounts payable	$ 100,000	$ 350,000	$ 110,000
Intercompany payable	40,000		
Amount owing to Z Co	50,000		
Share capital	1,000,000	500,000	200,000
Retained earnings	4,840,000	1,640,000	1,020,000
	$6,030,000	$2,490,000	$1,330,000

Additional information:

(a) On 1 November 20x5, Y Co sold inventory to P Co at a transfer price of $200,000, and recorded a gross profit of $60,000 on the sale. The subsequent resale of this batch of inventory is as follows:

Resold to third parties during 20x5	20%
Resold to third parties during 20x6	60%
In inventory as at 31 December 20x6	20%

(b) In 20x6, P Co sold excess inventory to Y Co as follows:

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Transfer price of inventory	$100,000
Original cost of inventory	$120,000
Percentage resold to third parties during 20x6	80%
Percentage in inventory as at 31 December 20x6	20%

The inventory was not impaired as at the date of transfer or subsequently.

(c) On 1 January 20x6, Z Co transferred an item of equipment to P Co at a transfer price of $172,000. The original cost of the equipment was $180,000 and the accumulated depreciation at the date of transfer was $36,000. The original useful life of the equipment was five years and the remaining useful life at the date of transfer was four years. Assume zero residual value.

(d) P Co recognized interest expense charged by Y Co and Z Co during 20x6 as follows:

Interest expense charged by Y Co	$1,500
Interest expense charged by Z Co	$1,000

The same amounts were recorded as interest income by Y Co and Z Co, respectively.

(e) Assume a tax rate of 20%. Recognize tax effects on fair value adjustments.

Required:

1. Prepare consolidation and equity accounting entries for the year ended 31 December 20x6, with narratives and workings.
2. Perform an analytical check on the balance in non-controlling interests and investment in associate as at 31 December 20x6, showing workings clearly.
3. Prepare the consolidation worksheets for the year ended 31 December 20x6.